Manual For

Screeners for the Alberta Universal Newborn Hearing Screening Project

This is a reference manual for Registered Nurses or other designated screeners in the province of Alberta and their role in the Alberta Universal Newborn Hearing Screening Project. The manual begins with an awareness of the need for universal newborn hearing screening programs. It provides an outline of the structure and protocol of the Alberta Universal Newborn Hearing Screening Project. It introduces screeners to the anatomy and physiology of infant hearing. Finally, a discussion of the current technologies as well as the successful administration and recording of infant hearing tests is also presented.

This project has been funded by the
Health Innovation Fund
Alberta Health and Wellness

Version 4.0
July 2002

Section 1

In this section you will:

- ✓ Understand the concept of and the need for universal newborn hearing screening programs.
- ✓ Comprehend the impact of a universal newborn hearing screening program for babies born in Alberta.
- ✓ Understand the objectives and structure of the Alberta Universal Newborn Hearing Screening Project.

What is Universal Newborn Hearing Screening?

Universal newborn hearing screening (UNHS) in an approach to the early identification of hearing loss in infants. It is designed to identify infants with the potential for hearing loss amidst a hospital population of newborns. Universal newborn hearing screening works in two stages. First, it utilizes a single or series of hearing tests designed with specific "pass" or "refer" criteria to identify those infants at risk for hearing loss. Secondly, the screening process ensures that these infants are diagnosed prior to three months of age and that they receive appropriate intervention or treatment by six months of age.

The Implications for Infants

Why is the early identification of infants born with hearing loss warranted? The following statistics have been available to health care providers for a number of years:

- Six out of every 1000 babies will be born with a hearing loss.
- The average age of detection of hearing loss in North America is 3 years of age. Lesser degrees of hearing loss will remain undiscovered even longer.
- The first 3 years of a child's life are critical for normal speech and language development.

The need for early detection of hearing loss was again emphasized by a recent, long-term research project by researchers at the University of Colorado.[1] The study revealed that children identified at birth with mild-to-severe hearing loss and who received intervention before 6 months could still fall within a normal range of language comprehension and expression as well as social development. In marked contrast, children with hearing loss diagnosed after six months of age experienced significant delays in both language and social development.

In the past, an infant's hearing may have been assessed by observing a startle response when the infant was exposed to sound. Some hospitals also maintained High Risk Registries where infants identified to be at risk for hearing loss, such as those in Neonatal Intensive Care Units, were tested for hearing. High Risk Registries can be inadequate however, missing as many as 50 % of infants with a hearing loss. In many parts of Canada, the current method of identification is the "wait and see" approach: parents, educators and/or physicians observing a growing child sense that their hearing is amiss and request an evaluation. The unfortunate result is a child with a hearing loss that has not been identified for a period of 18 months or longer. This in turn compromises the child's speech, language, cognitive and psychosocial skills as well as impacting their academic potential.

The intent of universal newborn hearing screening is to improve an individual's quality of life. It has already become a standard of practice in many European countries and numerous U.S. states. Alberta and Ontario will be the first Canadian provinces implementing programs for newborn hearing screening.

The Implications for Albertans

The percentage of infants born in Alberta, both currently and in the past, who have received a hearing screening, are very low.[2] In 1998, 37,779 infants were born in Alberta. Given that the prevalence rate of infants born with a hearing loss is 6/1000, approximately 227 infants should have been identified and diagnosed. Only 8 infants in the 0 to 3 years category were actually found. This is partly due to the fact that only two of the province's birthing hospitals offer screening programs. Furthermore, both sites confine their screening to the Special Care Nurseries (SCN) and Neonatal Intensive Care Units (NICU). This emphasis on high risk infants is also inadequate; in 1999 only 30% or 283 NICU infants at Foothills Hospital, Calgary, Alberta received a hearing screening.

On a national level, the opportunities for Canadian infants to receive a hearing screening again remain very low. Of the over 400 birthing hospitals in Canada, only 35 have some type of hearing screening program. If one examines the 1998 birth rate for Canada, 2233 infants were born with some degree of hearing loss amidst a total of 340,891 births. Identification and diagnosis of hearing loss remains very difficult.

The Alberta Universal Newborn Hearing Screening Project

The Alberta Universal Newborn Hearing Screening Project (Alberta UNHS Project) is funded by Alberta Health and Wellness for a trial period of 3 years. Beginning in the spring of 2001, it will develop and implement a universal newborn hearing screening program for the province's participating regional health authorities. The Alberta UNHS Project will identify infants with hearing loss by 3 months of age as well as ensure that they receive the appropriate follow-up and begin intervention before 6 months of age. This is a combined hospital and community-based project that will endeavor to service most of the province, while at the same time, reducing costs and saving valuable time in identifying those children with hearing loss. A model of the project and its interactions within the province and region is at the end of Section 1. The Alberta UNHS Project is based on models developed in other centres as well as the expertise gained by the investigators, University of Calgary, over the past 10 years.

Objectives of the Alberta UNHS Project

The principle objective of this project is to develop and implement a universal newborn hearing screening program identifying infants with hearing loss by three months of age. The regional health authorities (RHA) participating in the project are listed below. In order for this project to be successful, the following objectives must be met:

1) A minimum of 95% of all newborns will be screened by 3 months of age.
2) All or 100 % of all infants who do not pass the hearing screening are provided with adequate follow-up through a provincial-tracking program.
3) All or 100 % of hearing impaired infants will receive diagnosis of their hearing loss and be fitted with appropriate amplification by 6 months of age.

The project will also address the following issues:

1) The functional implementation of a UNHS program in each of the participating RHAs.
2) The modification of the program to specific RHAs taking into consideration individual regional issues or needs i.e. rural versus urban populations, cultural differences, etc.
3) Provision of hearing screening in a cost-effective manner.
4) Monitoring of outcomes and program efficacy.

Participating Regional Health Authorities

Region	1	Chinook Health Region
	2	Palliser Health Authority
	4	Calgary Regional Health Authority
	13	Mistahia Regional Health Authority

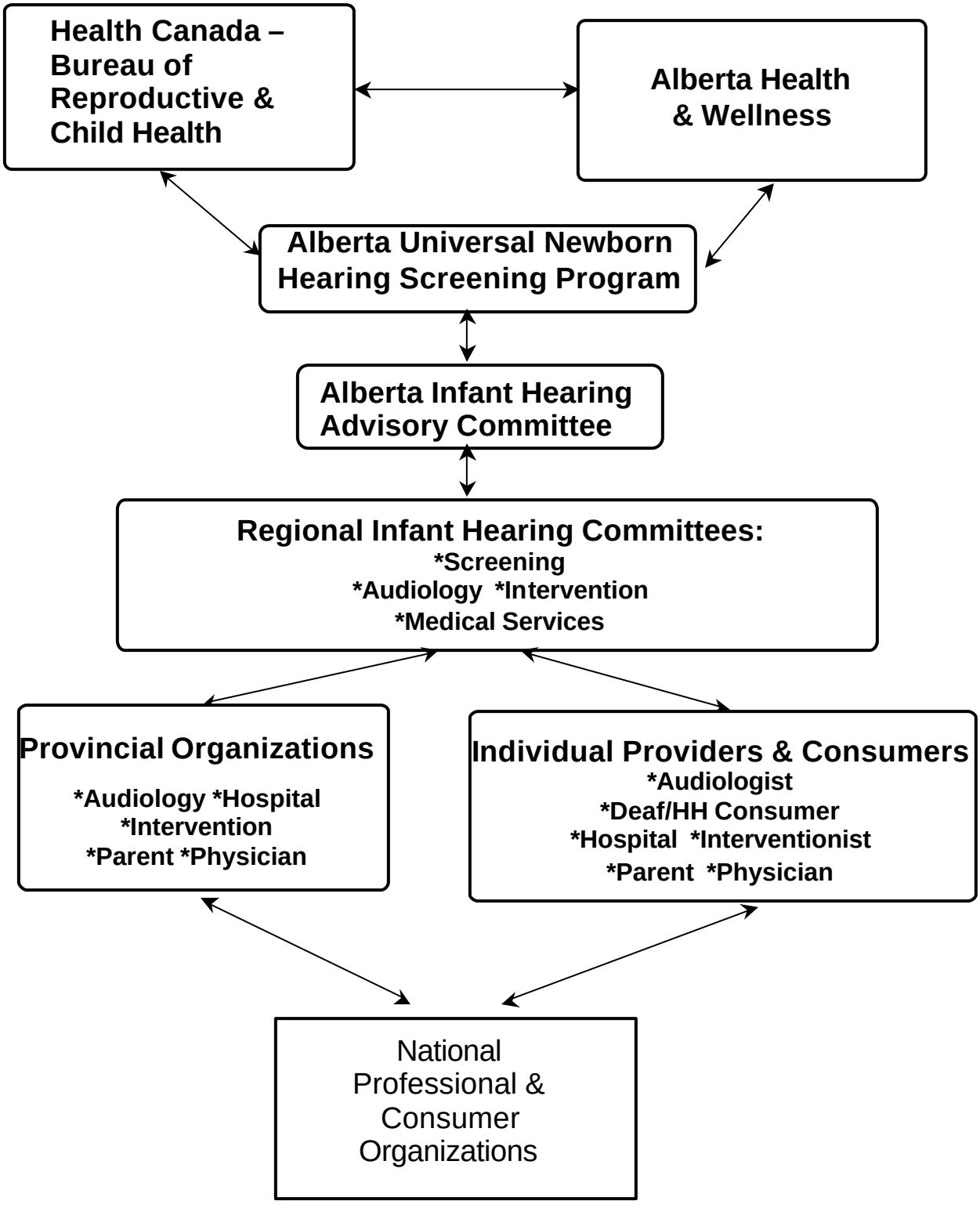
Stages of the Alberta UNHS Project

The Alberta UNHS Project will utilize a two-stage, two-technology approach. **Stage 1** will be a screening of all infants with a handheld Distortion Product Otoacoustic emission (DPOAE) unit. The hearing screening should occur within approximately 12 hours following a normal birth. Pre-term infants and those admitted to neonatal intensive care units and other step-down units will be screened according to different guidelines. These infants should be a minimum of 34 weeks gestational age, medically stable and as close to discharge as possible. Infants for whom the screening confirms normal hearing bilaterally will not require further testing.

If normal hearing is not confirmed, the infant will be identified or flagged for follow-up testing. This second screening will be done with the same DPOAE equipment within approximately 8 hours. If normal hearing is not confirmed on this second screening, the infant will be tested with AABR either in the hospital or in a regional outpatient facility. Use of AABR screening constitutes **Stage 2** of the screening process. If normal hearing remains unconfirmed by the AABR, the infant will then be referred for further audiometric evaluation. This technique is intended to provide the lowest or most accurate refer rate.

The audiological diagnostic assessment should be performed within one month of the above screenings. The regional audiology department will provide the diagnostic and habilitative requirements of the project. Treatment options will vary but should hearing aids or other types of amplification be required, they will be made available through the Alberta Aids to Daily Living program. Early Intervention will be initiated prior to 6 months of age.

Alberta UNHS Project Proposed Model



Section 2

In this section you will:

- ✓ Review the anatomy of the infant ear.
- ✓ Understand the physiology of hearing or why we hear.
- ✓ Differentiate between the different types of hearing loss.
- ✓ Recognize that some infants may have a higher risk for hearing loss than others.

Anatomy Of The Ear: Why We Hear

The human auditory system can be subdivided into the outer ear, middle ear and inner ear. These three basic structures and their various components allow us to hear.

Outer Ear

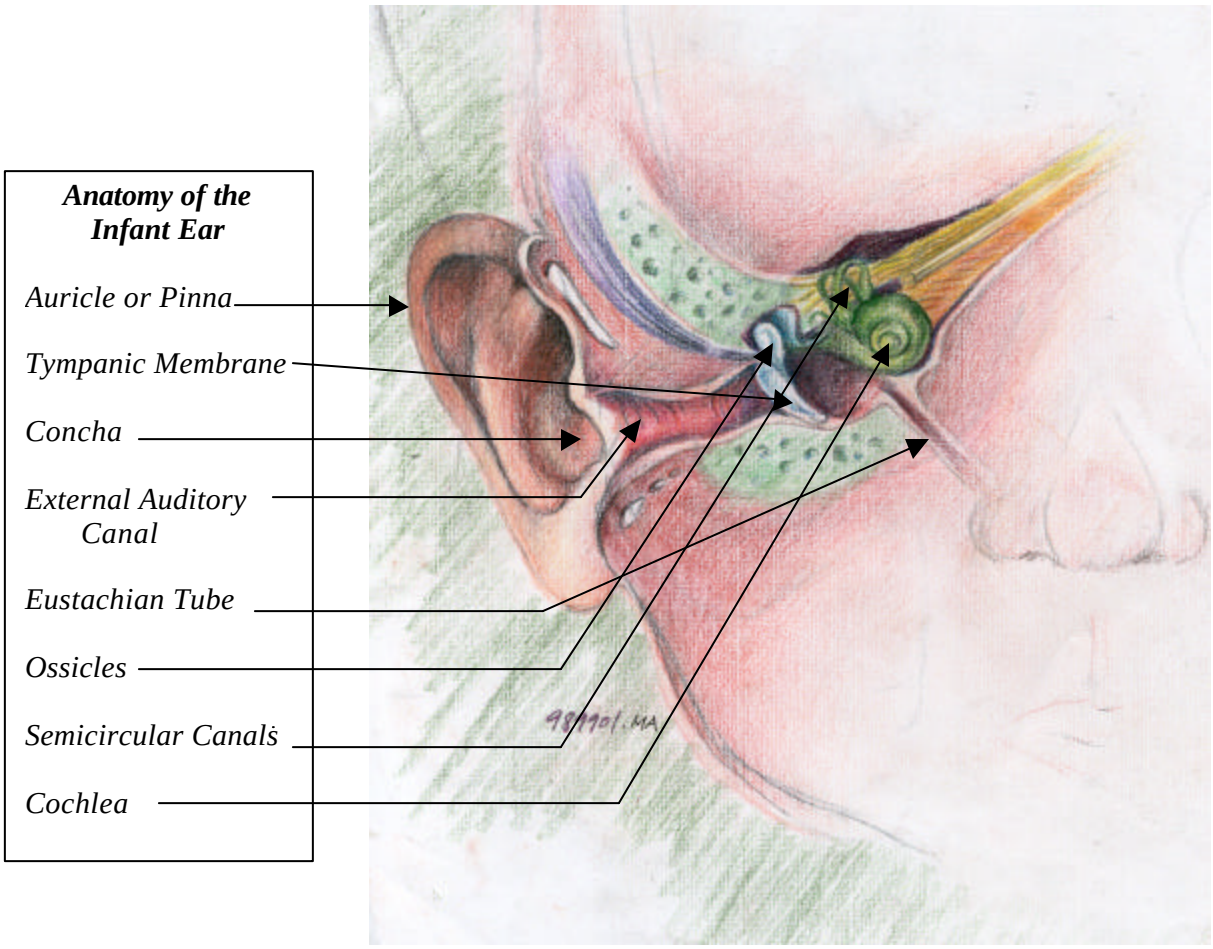
The outer ear, which includes the *auricle* or *pinna*, the *external auditory canal* and the *tympanic membrane* is the channel through which sound travels from the environment to the auditory system. The pinna is the most visual part of the ear and extends laterally from the side of the head. It is composed of cartilage and skin. The external auditory canal is a long, narrow tube beginning at the *concha*, the bowl-like portion of the pinna, and ending at the tympanic membrane. The canal is lined entirely of skin and is a resonant tube through which sound waves may pass. The tympanic membrane is a concave, disk-like structure which marks the common boundary between the outer and middle ear. It vibrates in reaction to the sound waves that strike it. It is sometimes referred to as the eardrum.

Middle Ear

The middle ear carries vibrations from the outer ear to the inner ear by transferring the sound energy from the air in the outer ear to the fluids of the inner ear. This is facilitated by three tiny bones called *ossicles*. The bones bear a Latin name describing their shape: *malleus* for hammer, *incus* for anvil and *stapes* for stirrup. Another important structure of the middle ear is the *eustachian tube* which connects the middle ear to the upperpart of the throat or the nasopharynx. The eustachian tube has two important functions. First, it supplies air to the middle ear which acts to equalize the air pressure on both sides of the tympanic membrane. This allows the tympanic membrane to vibrate effectively. Secondly, the eustachian tube allows for the drainage of fluids from the middle ear into the nasopharynx. In adults, the eustachian tube assumes a slight and downward slant. In infants and young children however, it lies horizontal to the nasopharynx, thereby facilitating the pooling or collection of fluid in the ear canal itself. This anatomical positioning can interfere with the passage of sound if fluid cannot drain normally and regularly from the area.

Inner Ear

The inner ear is a complex and physically very tiny structure found deep within the temporal bone of the skull. Since it is often compared to a intricately winding cave, the inner ear is called a labyrinth. The osseous labyrinth defines its bony outer casing and is divided into three additional sections: the *semicircular canals*, the *vestibule* and the *cochlea*. The semicircular canals and the vestibule contain the sensory organs for the vestibular system. The vestibular system is responsible for helping to maintain balance and posture. The coiled and snail-shaped cochlea contains the *organ of Corti*, which is the sensory organ for hearing. The organ of Corti contains several thousand sensory receptor cells called *hair cells*. Both the inner and outer hair cells are situated on the basilar membrane. Inner hair cells act as sensory processes by converting the mechanical energy produced by vibration into electrical energy. This is a form of passive mechanics. Active mechanics is required for soft sounds to be heard. Therefore energy must be added into the system through the motor processes of the outer hair cells. Messages are transported along the auditory nerve, or VIII cranial nerve, to the auditory cortex of the brain.



Types Of Hearing Loss

A) Conductive Hearing Loss

Conductive hearing loss occurs where some form of mechanical blockage impedes the flow of acoustic energy or sound to the inner ear. The blockage can be the result of birth debris such as vernix in the ear canal or amniotic fluid in the middle ear in babies less than one day old. Blockage however, is typically caused by fluid retention. This condition, known as otitis media (OM), is the most commonly diagnosed childhood disease between birth and 7 years of age.

Conductive hearing loss can also be caused by irregularities of the outer ear where the cartilaginous portion, the bony portion or the entire external auditory canal has never formed at all. This lack of canalization is called atresia and is a congenital abnormality. When there is an ear malformation in a young child, it is sometimes difficult to differentiate atresia from stenosis, which is a marked narrowing of the canal. Finally, a closed external auditory canal may be the result of trauma or burns. Trauma to the outer ear may result in an hematoma which can effectively block the passage of sound.

B) Sensorineural Hearing Loss

Sensorineural hearing loss is caused by damage or alternation of the cochlea or due to a problem further along the ascending neural pathways. For example, the hair cells may be damaged which

will impede electrical impulses normally originating in the hair cells from passing along the auditory nerve to the brain.
It is also possible to have a mixed hearing loss which is a sensorineural hearing loss with a superimposed conductive hearing loss.

C) Unilateral Versus Bilateral Hearing Loss
Unilateral hearing loss describes loss of hearing in one ear only. Bilateral hearing loss is loss of hearing in both ears.

D) Mild Versus Severe Hearing Loss
The degrees of hearing loss and the resulting repercussions or handicaps are described below.

Degrees of Hearing Loss

Category	**Probable Handicap**
Mild 27 to 40 dB HL	Faint or distant speech may be difficult to follow.
Moderate 41 to 55 dB HL	Conversational speech is understood up to a distance of 3-5 feet. As much as 50 % of class room discussions may be missed if voices are faint or not in the line of vision.
Moderately Severe 56 to 70 dB HL	Speech must be loud to be understood. Group discussions are difficult to follow.
Severe 71 to 90 dB HL	Voices may only be heard from a distance of about 1 foot from the ear. Environmental sounds and vowels may be discriminated but many consonants will be distorted.
Profound More than 91 dB HL	The individual may hear some loud sounds but is more aware of vibrations than tonal patterns. Vision rather than hearing is the main avenue for communication.[3]

Any form of hearing loss will impact a growing infant and child. This may also include chronic OM if a child is continually experiencing degrees of hearing loss because of this condition. Again, the repercussions may be wide-spread, and include cognitive, perceptual, speech, language and eventually, academic and social development of the child.

Some Babies Are More At Risk Than Others

Certain infants are known to be at risk for impaired hearing. This information is readily obtained from the birth mother's history as well as the infant's pre and postnatal assessments. It is important to observe that the following criteria should serve as a flag that an infant may be at risk; many of these babies can and do have normal hearing.

- A strong family history of infant or early childhood hearing loss. *
- The presence of cranio-facial abnormalities, i.e. ear tags, atresia, Down's Syndrome.
- High hyperbilirubinemia levels requiring at exchange transfusion level.
- A birth weight of less than 1500 grams.
- Congenital or prenatal infections including Cytomegalovirus, Syphilis, Herpes Simplex virus, Rubella, and HIV.
- Prolonged mechanical ventilation for a period of 5 days or longer.

*A strong family history is considered to be the most significant risk factor.

Section 3

In this section you will:

- ✓ Be introduced to two types of technology used to test hearing in infants.
- ✓ Learn how to prepare an infant for a hearing test.

Technology

Early identification of hearing loss can be accomplished through the use of either the auditory brainstem response (ABR) or otoacoustic emissions (OAE). Both objective measures have been shown to achieve the goals of an UNHS program. Many studies have shown that either test is a useful tool and that they are not only accurate but also fast, inexpensive and non-invasive. The two technologies are briefly discussed below.

Auditory Brainstem Response (ABR) is an electrophysiologic response that measures the auditory system's response to sound. A soft click is presented to the ear via earphones or a probe. Electrodes record the response as the sound travels from the ear through the auditory nervous system to the brain. Conventional ABR requires a trained technician or audiologist to use the ABR equipment. An audiologist is also required to interpret the test results.

Otoacoustic Emissions (OAE) is a screening method that measures the integrity of the outer hair cells in the inner ear or the cochlea. Either a pair of soft tones or a series of clicks is presented via a probe placed in the infant's ear canal. The infant's ear should respond by producing its own sounds that are detected by a small microphone within the probe. The response is analyzed to determine how well the cochlea is working.

The technologies used in newborn hearing screening have progressed in recent years. Current technology allows trained hospital personnel , such as nurses and technicians, to perform several screening procedures. The ABR and OAE equipment have become fully automated and no interpretation is required. The new screening devices illicit a PASS or REFER outcome.

Preparing The Baby

A hearing screening is best done on a captive audience such as newborn infants prior to discharge from hospital. Hearing screenings must be performed on sleeping or alert but quiet infants. Appropriately trained registered nurses can be very effective screeners; they are the immediate caregivers in hospital nurseries and have access to infants when they are in the correct state for testing. The following measures will help ensure an effective hearing screening of an infant:

- Is the infant available? Choose a time when the infant is not being seen by other health professionals or required for other tests. Wait until 12 hours after birth or following the infant's first bath which helps to remove birth debris.

- When was the infant last fed? Infants are often most content after a successful feeding. The ideal time is often approximately one hour after a feed. This will also prevent any regurgitation.

- Is the infant well? The infant should be medically stable. Avoid testing infants when they are febrile, congested or agitated. Speak with the primary care giver if the infant is in a Neonatal Intensive Care Unit.

- Where is the infant? A screening can be conducted in a nursery, at the bedside or any other room that is quiet. Background noise can effect test results and prolong the time required to complete the testing particularly with the otoacoustic emissions tests.

- Wash your hands thoroughly and clean the equipment for the testing before handling any infant.

- Is the infant in an appropriate position? The infant's entire ear must be clearly visualized and ideally facing upwards. Turn the infant on its side or gently turn its head. Observe the position and external appearance of the ear. Any external fluid or debris may be wiped clean.

- Is the infant in a quiet state? Swaddle and/or comfort a potentially restless baby. Dim the lighting. Have the baby held by a parent or by a staff member.

- Be familiar with the technical components of the screening. This will vary depending upon the type of test used but all probes must be properly positioned in the infant's ear canal. This will involve gently but firmly lifting the pinna upwards away from the infant's head and then pulling the pinna towards the back of the head. The ear canal should be clearly open. As one becomes proficient with conducting the tests, the amount of time required for their successful completion will decline. Emission tests on a quiet infant can be completed in less than one minute per ear.

- Report all results to the infant's parents. Enter the screening results in the appropriate hospital charting.

Repeat the screening if normal hearing is unconfirmed and there is time before the infant is discharged. At times, screening is successful when the infant is a few hours older or when fluid retained in the ear canal has had an opportunity to shift.

Additional Considerations for NICU Infants

If the infant is in a NICU or in a step-down unit requiring specialized care, it is best to test to hearing in the week prior to discharge; the infant will be most medically stable at this point. Other factors to consider when testing NICU infants is that they should be at least 34 weeks old in terms of gestational age. These parameters will ensure that the infant is physically and neurologically mature enough to undergo a hearing test. Always seek medical advice from the infant's immediate caregiver prior to performing a hearing test.

Section 4

In this section you will:

- ✓ Understand why the **ERO · SCAN** Distortion Product Otoacoustic Emission Test Instrument can effectively and accurately test hearing in infants.
- ✓ Identify the working components of the **ERO · SCAN**.
- ✓ Understand the need for good probe placement to obtain accurate test results.
- ✓ Perform a hearing test in the DPOAE Mode of the **ERO · SCAN**.
- ✓ Comprehend the "PASS" or "REFER" criteria of the **ERO · SCAN**.
- ✓ Be familiar with the cleaning and maintenance requirements of the **ERO · SCAN**.
- ✓ Be aware of trouble shooting techniques and correctly interpret error messages.

The Ero · Scan Distortion Product Otoacoutic Emission Test

What is the ERO · SCAN Instrument?

The **ERO · SCAN** Distortion Product Otoacoustic Emission (DPOAE) Test Instrument is a hand-held device designed to measure outer hair cell function through the measurement of DPOAEs. The name **ERO · SCAN** stands for Etymotic Research Otoacoustic Scanner. It has proven to be an effective screening tool in UNHS programs and will be extensively used in the Alberta UNHS project.

Physiological Background – What Are DPOAEs?

Distortion product otoacoustic emissions (DPOAEs) are acoustic signals generated by the cochlea's outer hair cells. When DPOAEs are detected in an infant, this is an indication that the outer hair cells of the cochlea are functioning normally. The **ERO · SCAN** instrument generates two soft tones at specific frequencies (f_1 and f_2), directs them into the infant's ear canal and then measures the response that is generated by the cochlea. A microphone in the probe measures and analyzes this response.

Since the **ERO · SCAN** measures the infant's response as well as the ambient and environmental noise, some sort of distinction must be made between the two. In other words, is the strength of the infant's cochlear response strong enough to rise above the noise floor that is generated by the infant's own breathing, the sound of an air conditioner, or the chatter of nearby individuals? Since ambient noise is generally more intense in the lower frequencies, the tones emitted by the **ERO · SCAN** are in the higher frequency range of 2000 Hz to 4000 Hz. The infant's response should be sufficiently strong to rise above the noise floor by a measurement of 5 dB or greater. We call this a Signal-to-Noise Ratio (SNR) of 5dB. Higher frequency DPOAEs are also used because the majority of significant hearing losses involve impairments in the high frequency range. Infants who fail to generate DPOAEs should be rescreened and/or referred for additional audiologic testing: please refer to the Alberta UNHS protocol.

THE ERO · SCAN AND ITS COMPONENTS





A box of eartips is provided with the **ERO · SCAN**.

The **ERO · SCAN** test device consists of a hand-held unit similar in size to an automated infant thermometer. It contains all the hardware and software for generating the test tones, measuring and displaying the DPOAEs as well as storing the results until they are printed. It includes an optional probe, single-use eartips and accessories.

A) The plastic housing features a front display as well as simple arrow keys to conduct and view the screening test.

B) Three indicator lights provide a visual display test conditions to the operator. They will flash the colours red, yellow and green to indicate if ideal conditions are present for a hearing screening.

C) A battery drawer slides open to reveal the compartment for the 4 AA batteries required to power the **ERO · SCAN**. Slide the battery panel down and install the batteries as indicated on the battery label. Once the batteries are correctly in place, slide the panel back onto its tracks and close the battery compartment.

D) The probe section or optional corded probe houses a microphone and two speaker tubs which produce the two sounds (f1 and f2) and which measure the sound pressure levels (SPL) present in the occluded ear canal. The microphone is able to measure the infant's cochlear response. Disposable eartips fit onto the probe tip and are available in several sizes for a correct fit of the infant's ear.

Probe Placement

Infants and their ear canals come in all shapes and sizes. The ear tips designed for **ERO · SCAN** come in a variety of sizes and are disposable after use on each individual baby. The determination of the appropriate size is simple and becomes readily obvious after a few screenings. Eartip sizes for infants typically range from 4 mm to 5 mm with 4 mm being the smallest size. The smaller infants usually found in NICU's are usually fitted with a 4 mm eartip. Full-term infants in well-baby nurseries are generally larger infants and are most successfully screened with a 5 mm eartip and occasionally, a 6 mm eartip.

After selecting an eartip, push it onto the probe tip until it is flush against the base of the probe tip. The sound outlet tubes on the probe tip are recessed to help with their becoming clogged with birth debris or wax. If the probe tip does become plugged or clogged, this can easily be visualized by simply examining the eartip. The clogged eartip must be replaced with a clean one. Remove the eartip by grasping the tip at its base and gently pulling it off the end of the probe tip.

Additional supplies of eartips may be obtained from your regional audiologist. Use only the eartips approved for use with the instrument.

Occasionally an infant will have large amounts of vernix or other birth debris in its ears. An audiologist or physician should be asked to help remove the blockage before the hearing screening is performed. Excessive vernix may produce invalid or incomplete results. Other than gently wiping away any external debris from the ear, Registered Nurses and other screeners should not attempt cleaning the infant's ears if a blockage is suspected.

For replacement of the entire probe tip, please refer to **Probe Tip Replacement** in this section of this manual.



The ERO·SCAN Instrument probe tip must not be inserted into an ear at any time without a disposable eartip properly affixed.

Performing A Test With The Ero · Scan

Control Buttons
The ERO·SCAN uses the four buttons with cursors to direct the functions of the device. The arrows on the keys such as the ? right, ? left, ? up, ? down correspond to the arrows that are used on the screen. The screen will indicate which button to push by showing the appropriate arrow.



Turning On the Instrument
To turn on the **ERO · SCAN**, press the large ? down key located below the display screen. The indicator lights will briefly flash red, yellow and green. The green ready light will remain on. *Display #1* will quickly appear.

Note: The **ERO · SCAN** has an automatic "shut down" feature where it will automatically turn itself off after 60 seconds of inactivity. This feature is designed to prolong battery life. To turn the **ERO · SCAN** back on, simply press the large ? down key again.



Display #1

The **ERO · SCAN** will automatically provide the date and time settings. If there are no date/time errors detected, the Main Menu indicated in *Display #2* will appear on the display. The date and time settings are easily changed. Instructions for this may be found under the **Setting the Clock** in this section.



Display #2

Testing in the DPOAE Mode

The **ERO · SCAN** can use a DPOAE protocol in addition to other customized protocols to test for hearing. The Alberta UNHS project will utilize the DPOAE Mode to confirm hearing in infants.

The settings for testing in the DPOAE Mode are as follows:

Number of Frequencies Tested = 3
Frequency Range = 2 kHz to 4 kHz
P1 (Intensity of F1 in dB SPL) = 65
P2 (Intensity of F2 in dB SPL) = 55
Averaging Time = 4 Seconds
Pass SNR (Signal-to-Noise Ration) = 5 dB
Number of Passing Frequencies for an Overall Test PASS = 3

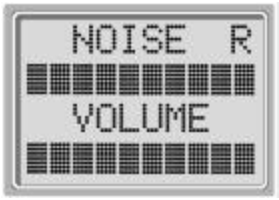
The **Advanced Options** menu allows the operator to change test or screening protocols and create other custom protocols. Registered Nurses screening for the Alberta UNHS project will always use the DPOAE Mode which will be automatically programmed for their use.

To Begin Testing

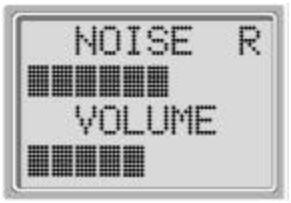
Begin the hearing screening by placing an eartip as far down as possible on the probe tip. Turn on the **ERO · SCAN** by pressing the down button ?and then selecting the left ? key or right ?key depending on which ear is being screened. Press the left ? if the left ear is being tested or press the right ? is the right ear is being tested. *Display #3* will appear on the display. This display shows two horizontal bar graphs representing the environment noise (NOISE) and the ear canal volume (VOLUME). When both bars fill the screen from left to right, the environmental noise is high and the ear canal volume is large. As the unit is inserted into the ear canal, both horizontal bars should decrease from right to left indicating a decrease in environmental noise and ear canal volume.

A good seal is essential to obtain accurate DPOAE measurements. The probe must be properly positioned in the ear canal. This will involve gently but firmly lifting the pinna upwards away from the infant's head and then pulling the pinna towards the back of the head. The ear canal should be clearly open. Gently insert the eartip into the ear canal. It should fit snugly and comfortably. The best test results are obtained when an eartip is inserted down into the ear canal instead of being flush with the ear canal opening. The NOISE and VOLUME bar readings should be used as an indication that the unit is sealing the ear canal. If the NOISE and VOLUME bars continue to fill the screen, the unit is not sealing the ear canal and appropriate adjustments should be made until a decrease in these bars is seen. *Display #3* illustrates how the **ERO · SCAN**

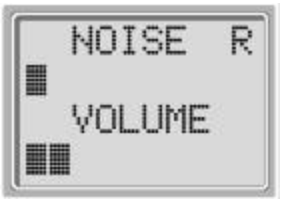
guides the screener towards obtaining a good seal. Once the positioning of the probe is correct, the **ERO · SCAN** will automatically begin testing. The yellow TEST light in the indicator light panel will stay on throughout the duration of the test.



Calibration will not begin since it is both too noisy and the probe positioning is poor.



Conditions for testing have improved but noise levels remain high.



Conditions have been met for calibration to begin.

Display #3

The Calibration Process

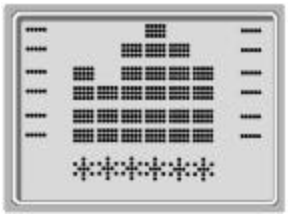
Prior to the start of the hearing test, the **ERO · SCAN** will automatically perform a **calibration.** *Display #4* will appear on the display. During the calibration process, a series of tones are presented to the ear canal to measure the levels of the primary tones at the frequencies to be tested. It is very important that the **ERO · SCAN** and/or remote probe be held steadily while the yellow TEST indicator light is on. Any excessive movement whether it be caused by the unsteadiness of the operator's hand or by the accidental movement of the infant's head, will break the seal in the ear canal. An error message will be displayed and the screening test will not continue.



Display #4

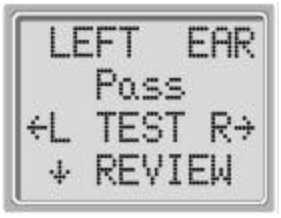
Bar Graph Display of the Test Results

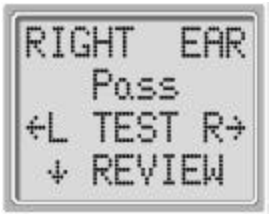
If the calibration process has been successful, the **ERO · SCAN** will present soft tones into the infant's ear canal. *Display #5* depicts the set of bar graphs that should appear on the display. The bar graphs are a visual presentation of the strength of the cochlear emissions. Each of the measured frequencies, obtained at 2000, 3000 and 4000 Hz, are represented by two bars. The screening test is complete when the green READY indicator light is on. Please refer to the section on **Interpretation of Test Results** in this section for explanations of the bar graph display.



Display #5

The PASS or REFER Display
When the hearing test is completed, *Display #6* will appear. This screen verifies the tested ear and indicates the results of the test. A PASS on the screen confirms hearing in the infant. A REFER indicates that normal hearing cannot be confirmed. The results of the hearing test are automatically saved in **ERO · SCAN**'s memory. If the unit is turned off when the batteries are temporarily removed, test results are still obtainable.
To review the results, push the ? down arrow key and the bar graphs will reappear. Push the ? down key a second time and the main menu will be displayed.





Display #6

Interpreting Test Results

Understanding the Bar Graph Display
The **ERO · SCAN** display is easily interpreted for the operator's convenience. The display will hold a 6 column graph as shown below. In the DPOAE Mode where only three frequencies are tested, two columns represent each frequency. Each row represents 2 dB of signal-to-noise ratio (SNR). Therefore, if all 6 rows go to the top of the display, then the SNR is 12 dB or greater. Recall that an infant will achieve a pass on the DPOAE emissions test if a SNR of 5 dB or greater is present on each of the three frequencies.

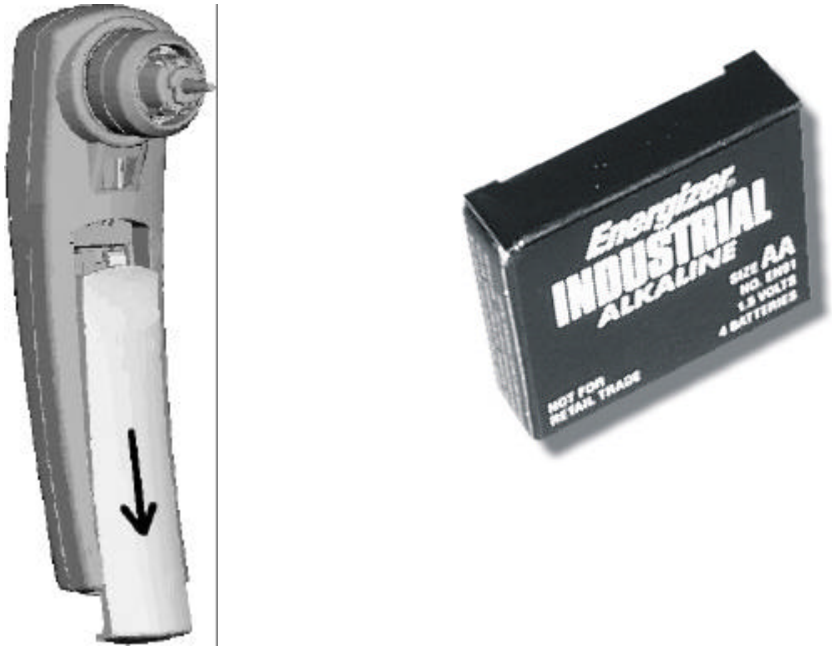
The display is shown while the test is being performed. The screener may also review the display after the test is completed by pressing the ? REVIEW key to look at the display again.

Cleaning With a Mild Antiseptic Solution
The **ERO · SCAN** and its accessories may be wiped clean with a mild antiseptic solution. which has been applied to a cloth. Clean the device prior to testing each infant. Only small amounts of cleaning solution are required. Fluid should not enter the device in any manner since this will interfere with the proper functioning of the **ERO · SCAN**. Likewise, the device should never be immersed into any fluid nor should the **ERO · SCAN**. or its accessories be sterilized.

Take care not to put excessive pressures on the clear display window or allow any utensil to puncture the display window or keypad.

Maintenance

The **ERO · SCAN** requires very little maintenance. It should be cleansed between tests as described above. Battery replacement will be required from time-to-time depending on usage. A battery drawer slides open to reveal the compartment for the 4 AA batteries required to power the **ERO · SCAN**. Slide the battery panel down and install the batteries as indicated on the battery label. Once the batteries are correctly in place, slide the panel back onto its tracks and close the battery compartment.



*Replacing batteries in the **ERO · SCAN** simply involves sliding the drawer out of the back of the intrument and inserting 4 AA Alkaline batteries.*

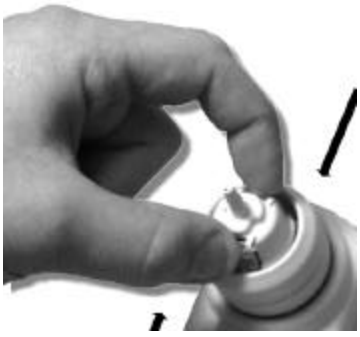
Recall that the **ERO · SCAN** has an automatic "shut down"feature where the device will turn itself off after 60 seconds of inactivity In addition, the ERO·SCAN can be turned off by pressing the ? button. Simply press the large ? down key to turn the ERO·SCAN back on and begin testing.

Probe Tip Replacement

The probe tip can occasionally become clogged with debris such as vernix or wax. The probe tip can sometimes be cleansed with a clean cloth if the debris appears to be minimal or restricted to the surface. If the debris appears to be embedded in the probe however, do not attempt to clean the probe since this is an awkward and often ineffective process. It should be replaced with a replacement probe tip included with the **ERO · SCAN** kit

***Removing the Probe Tip.**

- Using a small pointed object, such as a pen or small screwdriver, push in the notches on the left and right sides of the rear of the remote probe until each tab is released. Please refer to the figure below.



- Slide the probe tip off the front of the probe and discard.

Replacing a New Probe Tip:

- Align a replacement tip with the front of the probe. **The tip will only fit in one direction.** If the tip does not fit securely on the probe, remove the probe tip and reorient it.
- Press lightly on the tabs to snap them into place. The tabs should be flush with the outside of the probe. Please refer to the figure below.



Note: If the probe tip is not inserted completely, the ERO · SCAN will not perform a test.

Setting the Clock

When the **ERO·SCAN** test instrument is first used, the correct date and time will need to be set on its internal clock. The date and time are listed on the test printout as month-day-year (i.e. Feb-26-01). The clock should be set prior to testing, as changing it after tests are saved will not change the date on the printout (i.e. whatever date was previously in memory will be the date on the printout). **Seasonal time changes, such as Daylight Savings Time, will also require resetting the clock.**



Display #7

If the instrument's batteries are depleted and not replaced within approximately one hour, *Display #7* as shown on the left may appear. If this message appears, reset the time and date. To change the time and date, press ? SETUP at the main menu (*Display #2* in this section).



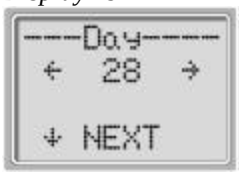
Display #8

Display #8 will show the current date and time presently set in the unit. If the time and place are correct, press the ? UP key to return to the main menu. If either the date or time is incorrect, press the ? CHANGE key to access the menu to change the month (*Display #9*).



Display #9

Press the ? LEFT or ? RIGHT keys to scroll forward or backward through the months. You will see the abbreviated name for each month.



Display #10

When the desired month appears on the display, press the ? DOWN key to enter the day selection screen (*Display #10*).



Display #11

Pressing the ? LEFT or ? RIGHT keys will scroll through the years (*Display #11*).



Display #12

When the correct year is selected, press the ? DOWN arrow key to enter the hour menu (*Display #12*). Again, use the ? LEFT or ? RIGHT arrow keys to set the correct hour.



Display #13

Press the ? DOWN arrow key to enter the minute menu (*Display #13*) and use the ? LEFT or ? RIGHT arrows to select the correct minute. When the correct minute is selected, pressing the ? DOWN arrow key will return you to the Main Menu.

The time and date changes are automatically saved.

Trouble Shooting

1. Problem: The ERO · SCAN does not turn on.

Solutions:
1) The ? down key must be pressed down for a full second.
2) Check that the batteries are installed correctly. There is a diagram for this on the back of the battery panel.
3) Try installing new batteries.
4) Check to see that the probe cable is properly attached.

2. Problem: The test will not begin even thought the probe has been placed in the infant's ear canal.

Solutions:
1) Verify that the eartip is sealed in the ear canal. Straighten out the ear canal by gently lifting the pinna upwards away from the infant's head and then pulling the pinna towards the back of the head.
2) Try a different sized ear tip.
3) Replace the probe tip since it may contain debris that is not visible.

3. Problem: The noise indicator light flashes continuously and the test will not begin.

Solutions:
1) If the source of the noise is external ie. loud, nearby conversation, inform the involved individuals that you are trying to perform a hearing test. If the noise is due to necessary monitors, fans, etc, a quieter area may need to be found. Do not disconnect the infant from any monitors or equipment; test the infant at another time under better conditions if possible.
2) If the noise is coming from the infant i.e. he/she may be breathing loudly, uttering sounds or even making sucking noises, attempt to soothe the infant. Ask a staff member or parent to hold the infant to comfort them. The infant may be hungry and need to be fed prior to its hearing test. If the noise is the result of nasal congestion, simply retest the infant at another time.
3) Remember that conducting a test under noisy conditions will produce unreliable test results even if the **ERO · SCAN** begins a test sequence. It is still advisable to test the infant at another time.

Section 5

In this section you will:

- ✓ Understand why the physiology of auditory evoked potentials and how they are interpreted with the **ABaer** hearing screening device.
- ✓ Identify the working components of the **ABaer**.
- ✓ Understand the techniques of good skin preparation and electrode placement for accurate test results.
- ✓ Perform a hearing test using the **ABaer**.
- ✓ Interpret the results of the **ABaer** test.
- ✓ Be familiar with the cleaning and maintenance requirements of the **ABaer**.
- ✓ Be aware of trouble shooting techniques with the **ABaer** device.

The Automated Auditory Brainstem Response (AABR)

Physiological Background

As discussed in the section on the anatomy of the ear, the auditory pathway is complex and consists of several stages. Sound cannot travel directly from the environment to the brain. By the time "sound" leaves the cochlea or inner ear, it has become a series of neuroelectric events travelling in sequence to the acoustic nerve, the brainstem, and finally to the cortical areas of the brain where it is interpreted. It is logical therefore that whenever a sound is heard, there are changes in the ongoing electrical activity of the brain. These electrical responses have been referred to by several names but they are now commonly known as **auditory evoked potentials**.

Measurement of auditory evoked potentials is possible through Auditory Brainstem Response (ABR) Audiometry. Three electrodes are positioned on the infant's head with their exact placement being determined by the type of ABR used. The electrodes are able to discern the electrical activity generated at all levels of the auditory pathway beginning at the inner ear and ending at the auditory cortex. Since the brain has all sorts of electrical activity going on concurrently, it is essential to isolate the auditory response. Any muscle activity, such as that brought on by yawning or the wrinkling of the forehead, produces its own electrical activity and can obscure the auditory response. Thus the infant is tested when sleeping or awake but in a quiet state. ABR also utilizes averaging computers which are devices that allow for the measurement of auditory evoked potentials even when they are embedded in other electrical activity.

One ear is tested at a time. A probe is placed in the infant's ear and a series of 1000 to 2000 soft clicks are presented at a very rapid rate, approximately 33.1 clicks per second. The starting intensity level of the clicks is often 70 dB which is equivalent to the noise level attained in a normal conversation. The electrodes will record the neuroelectric activity at one or more generating sites along the auditory brainstem pathway. A waveform with a series of upclines, peaks and declines will present themselves on a computer screen. The following is a simplified scheme of which waveforms correspond to the activity generated along the auditory pathway:

Wave Number	Site
I	VIIIth cranial nerve
II	VIIIth cranial nerve
III	Pons
IV	Pons
V	Midbrain

Since wave V has the largest amplitude or highest peak as well as being the most reliable of the responses, its presence is considered to be the most significant. After the test has been completed, a hard copy may be printed out where the data is plotted and summarized. A PASS or REFER judgement is automatically made.

Description of the Automated Brainstem Auditory Evoked Response Device: The ABaer

This is a computer-based system that performs an automated brainstem auditory evoked response (ABaer) screening. This screening device will identify infants who may have peripheral hearing

loss and who may require additional testing. It is designed for infants with a gestational age of 34 weeks through six months of age. The **ABaer** system automatically evaluates the ABR response with the Point Optimized Variance Ratio (POVR) signal detection algorithm. After a test, the **ABaer** system recommends a PASS or REFER by comparing the POVR with a preset criterion.
The **ABaer** software will be loaded on a computer. For computer specifications, please see the User's and Service Manual provided by Bio-logic Systems Corp. and which is available from the Regional Audiology Coordinator.
ABaer is equipped with two cables. The electrode cable is used to attach the electrode leads with appropriately labeled connection sites. The patient cable will have a probe on which an ear tip is inserted. Disposable electrode kits are required for the test as well as appropriately sized infant probe tips.

Skin Preparation and Electrode Placement

The skin at the electrode sites must be prepared to ensure a good recording. The three electrode sites are the high forehead and the mastoid bone behind each ear. Use 2x2 gauze wipes and the provided Nuprep gel to clean each of the areas. The gel is mildly abrasive. Swipe across the site with 3 to 4 moderately firm strokes in order to obtain a satisfactory impedance rating. The swipes should still be gentle enough to neither awaken nor harm the infant. Screeners will soon recognize how much swiping or cleaning is required to properly prepare the site. As with the otoacoustic emissions test, the **ABaer** test is most successful when performed on a sleeping infant. If the infant is agitated or active, the test may be prolonged or the results may be compromised by a high level of background noise in the recording.

Once the three areas have been cleaned, the disposable electrodes may be gently applied to the site. The electrode cable has both labels and colour coding to ensure that the electrodes leads are correctly connected.

If the electrode impedance is higher than is allowed, the program will automatically display a screen that shows the unacceptable electrode site(s) in red. It will be necessary to remove the electrode at that site and prep the skin a second time with the Nuprep. The program will continuously check the impedance values at each site and will automatically proceed to the screening test phase when the impedance at all three sites is acceptable.

Ear Probe Placement

Proper placement of the ear tip is important to ensure adequate delivery of the click stimulus to the ear. Tips are available in two sizes with a smaller size for premature and typically smaller infants and a " tree" or ridged tip for larger infants. It is important to verify that the placement of the tip and its transducer is accurate so that when a right ear test is specified in the program, it is the right ear that is actually receiving the click stimulus. Incorrect placement of the probe can result in poor quality ABR recordings, increase the chance that a REFER result will occur and assign the test result to the incorrect ear.

Performing a Hearing Screening Using the ABaer

The infant should be tested in a quiet area. It is essential that the area also be as free of electrically generating devices as possible. Devices such as portable x-ray machines can potentially generate their own electrical waveforms and thereby negate the results of the **ABaer**. This is called electrical interference.

Data Collection Parameters

The stimulus and recording parameters used for the **ABaer** test are listed below. The parameters are optimized for the screening technology that is being used by the **ABaer** system.

Stimulus Type	100 microsecond click
Stimulus Polarity	Alternating
Stimulus Rate	37.1
Stimulus Intensity	35 dB nHL (default)
Analysis Window	21.33 msec
High Pass Filter	100 HZ
Low Pass Filter	1503 Hz
Amplifier Gain	100,000
Channels	1
Montage	High forehead (active or non-inverting) to Test Ear (reference or inverting). Non-test ear is the ground.
Minimum Sweeps	1024
Maximum Sweeps	12,288 (in two separate trials of 6144 sweeps each).

How To Start The Abaer Program

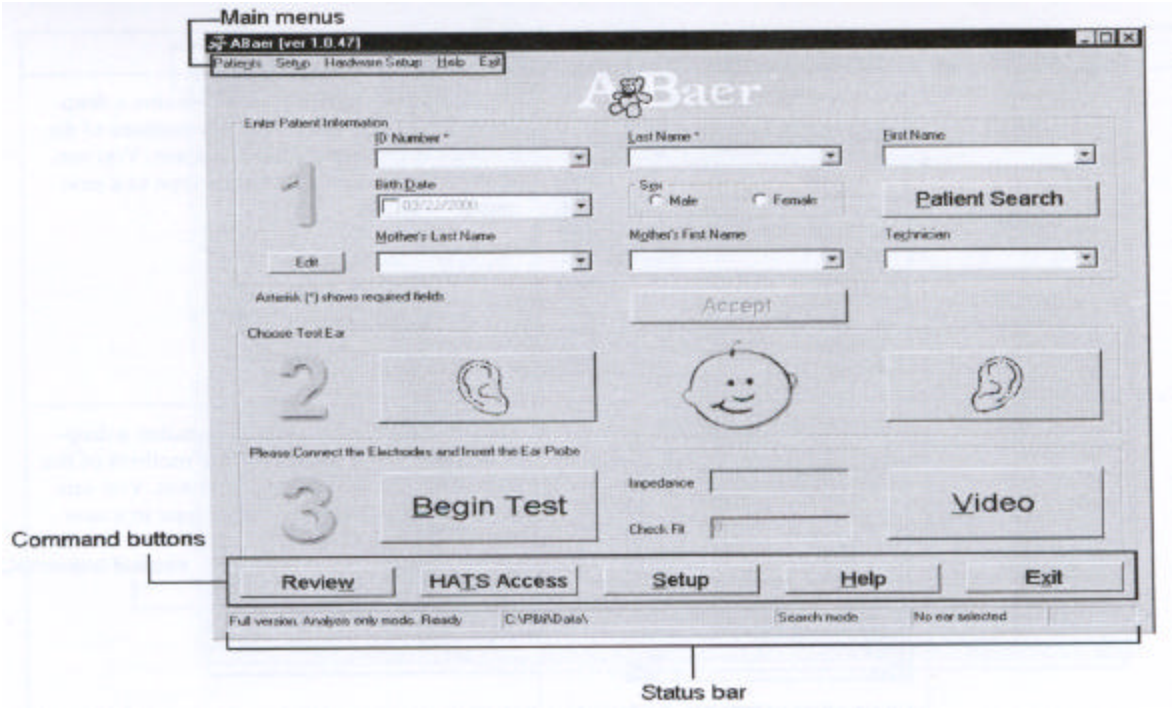
1. Activate the computer or turn on the computer and wait until the Windows "Desktop" screen appears, followed by the ABaer program.

2. If the ABaer program does not automatically appear, select (click on) the ABaer icon to start the program.

3. The following window will appear. Enter Patient information as follows:
   Place the mouse cursor in the ID Number field. Type in a new ID Number. This will be the infant's Unique Lifetime Identifier (ULI) number.

OR

Click on the down arrow in the ID Number field. A drop-down list of ULI numbers appears.
-Select the ULI number you want from the list.



4. Place the mouse cursor in the last name field. Type in a new last name.

OR
Click on the down arrow in the Last Name field. A drop-down list of last names appears. Select the last name you want from the list.

5. Place the mouse cursor in the First name field.

6. Type in a new first name.

7. Click on the checkbox in the birth Date field until a checkmark appears to accept the default date that appears in that field. This date usually appears in gray for a new test and is the date that was last entered in that field.

OR

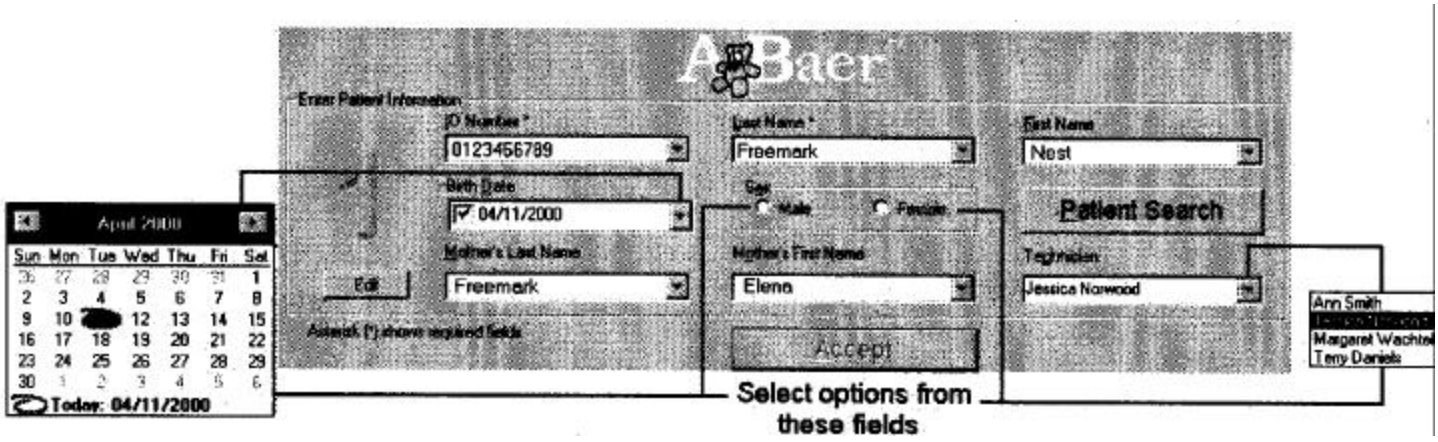
8. Click on the down arrow in the Birth Date field to activate a Calendar tool. Use the calendar tool as follows:
   Click on the increment (?) or decrement (?) buttons at the top of the calendar page to display the month and year you want to use. Click on the day you want to use.

9. Click on the radio button that reflects the gender of the infant you are testing. A bullet will appear.

10. Verify that the name that appears in the Mother's Last Name field is correct. If not, place the mouse cursor in the Mother's Last Name field and type in a new last name.

11. Place the mouse cursor in the Mother's First Name field and type in a new first name.

12. Click on the down arrow in the Technician field. A drop-down list of the names of the technicians who run the **ABaer** tests for your facility appears. –Select the name you want from the list.



OR
-Place the mouse cursor in the Technician field and type in a new name.

Note: The Abaer program adds this value to the list of names so that it appears in the drop-down list for future patients.

13. Check the information on the Step 1 area of the **ABaer** Main Screen.

14. Select or click on the Edit command button to change any information.

OR

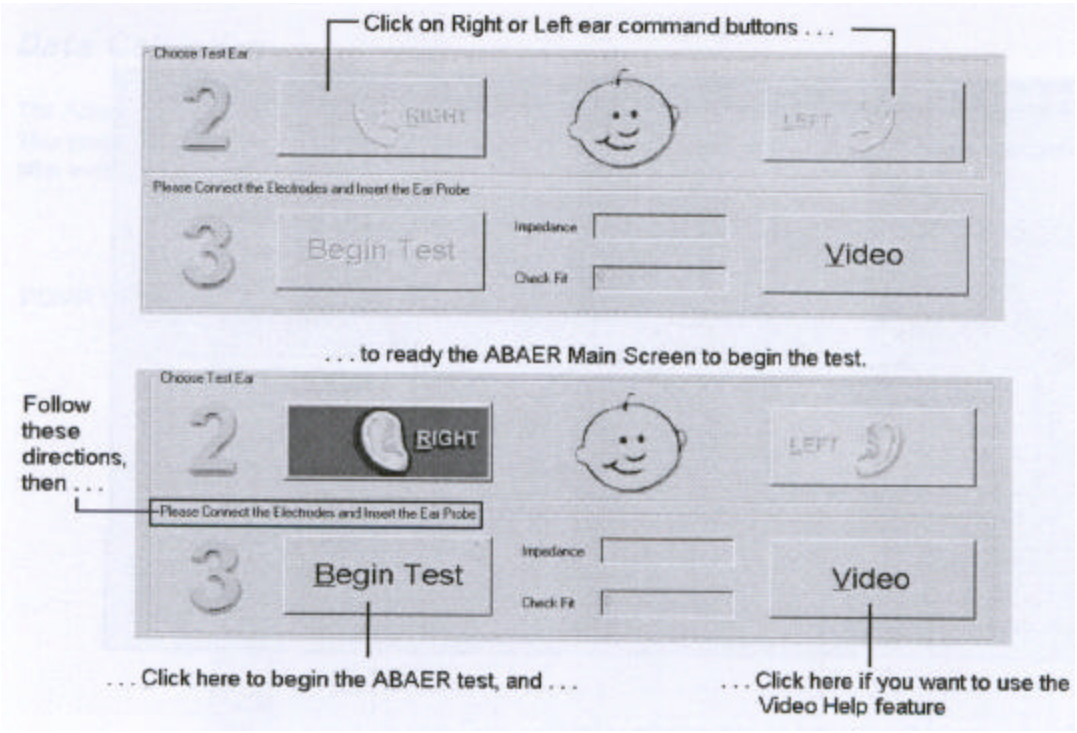
15. Click on the down arrow in any field and select the Clear All option. This erases all of the information in Step 1 and lets you start over.

OR

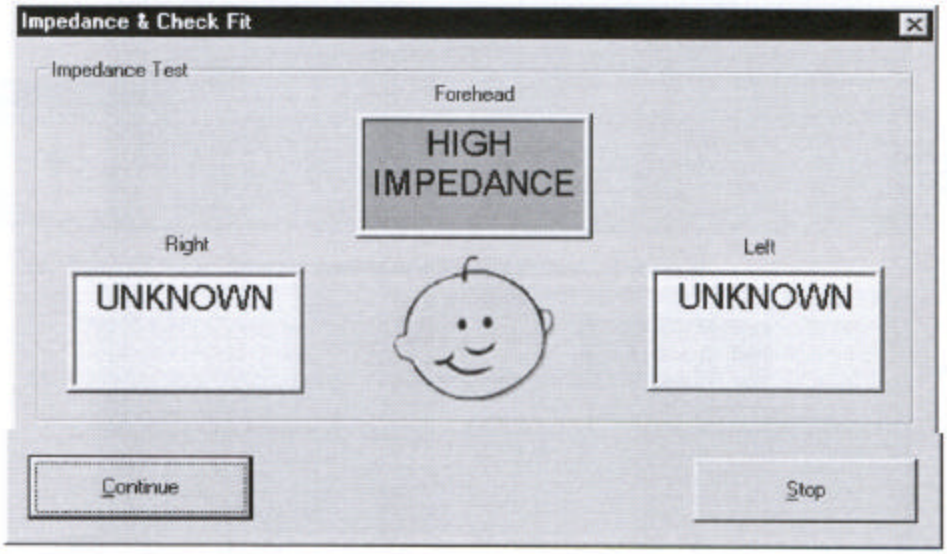
16. Click on the ACCEPT command button to activate the Step 2 area of the **ABaer** Main Screen. Please see the accompanying diagram.
17. Select or click on one of the buttons with the ear image. When you select an ear, the colour of the button changes from pink to red for the right ear and from light blue to bright blue for the left ear.

18. Connect the electrodes and insert the ear probe. Please see the section on baby preparation.

19. Select or click on the Begin Test button.



This initiates an impedance check. If the impedance values for all three electrodes are within the limits you defined for the test, then the **ABaer** program will automatically go to the date data collection phase. If the impedance for any of the three electrodes is higher than the acceptable limit, a secondary display appears.



This display shows the status of the impedance for each electrode site. The electrode sites with impedance values above the allowable limit appear in red and the electrode sites with acceptable impedance values appear in green. If the system cannot read impedance at a site, the system displays the area in yellow and labels it "Unknown."

20. If required, rescrub the sites that exceed impedance levels.

21. Consult the Impedance and Checkfit screen to see if the values improve.

OR

22. Continue the test even though the impedance criterion is not met.

Data collection begins. A new display that shows a graphical representation of the analysis of the ABR signal appears. After the system delivers each set of 245 stimuli to the ear, a data point appears on the graph. The value assigned to this assessment is the POVR score-a statistical measure of signal to noise in the ABR response.

The program follows a routine based on continuous comparison of the POVR measurement with a set of criteria that will automatically terminate the test and assign a Pass or Refer recommendation. The criteria for termination of the test are as follows.

- A POVR score of 3.5 achieved after a minimum of 1024 stimuli has been averaged.

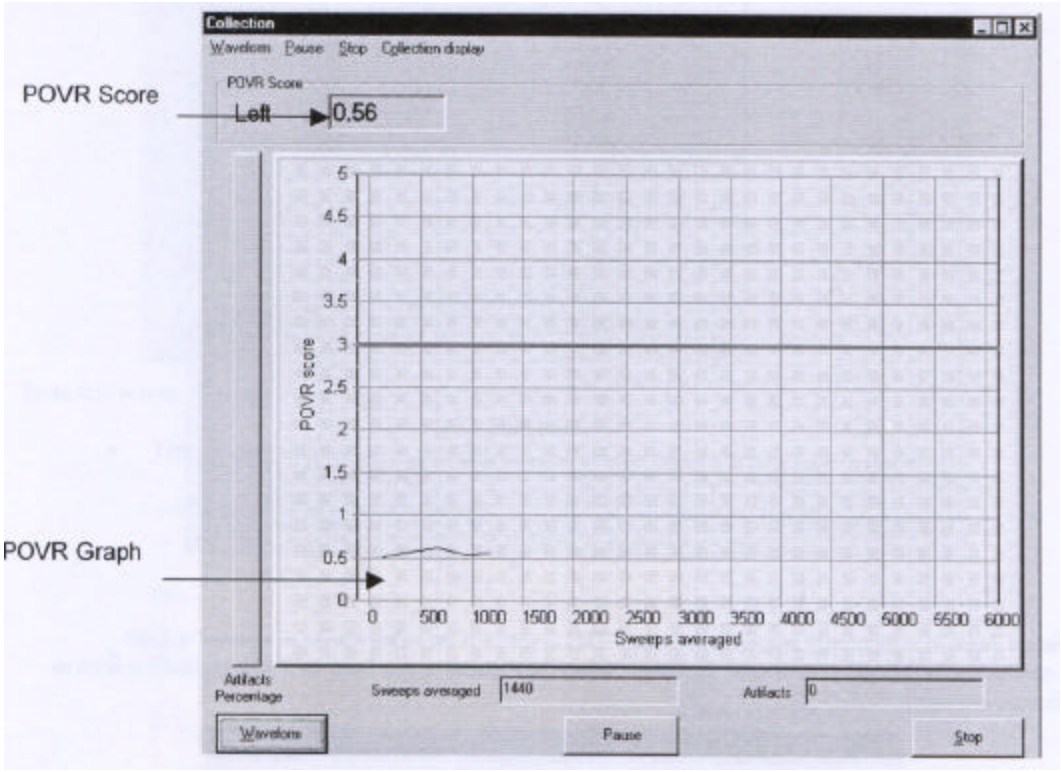
- A total of 6144 stimuli are averaged and the POVR score reached a value of 3.1 or higher.

- A POVR score did not reach or remain at a value of 3.1 at the end of two sets of 6144 stimuli.

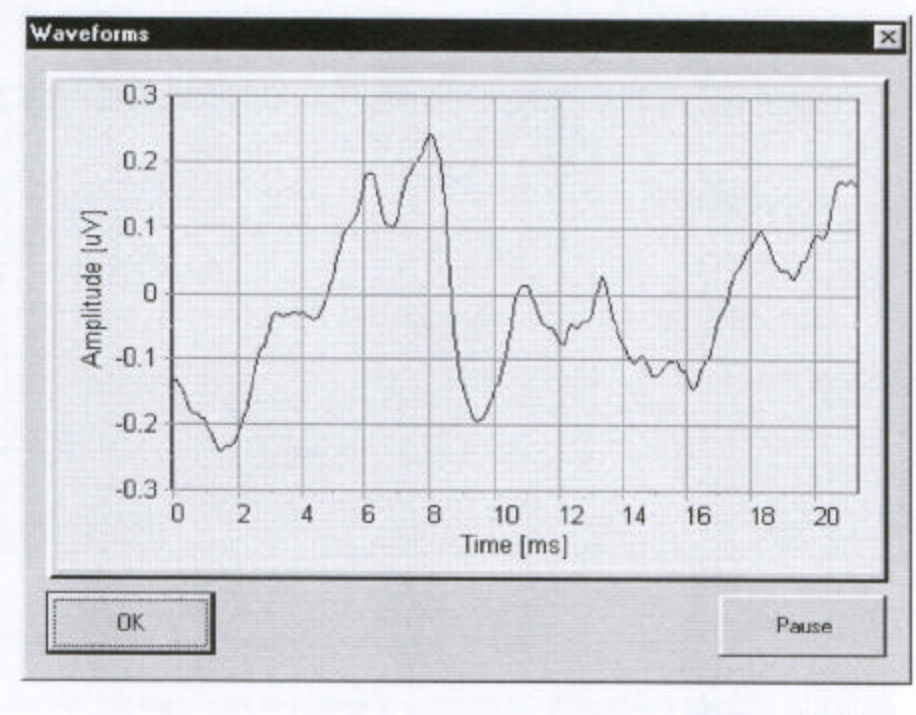
The system displays a PASS for a POVR score of 3.1 or higher after a minimum of 1024 stimuli. The system displays a REFER when the POVR score fails to remain at a value of 3.1 after the system averages 6144 stimuli twice.

Interpreting the Results of the AABR

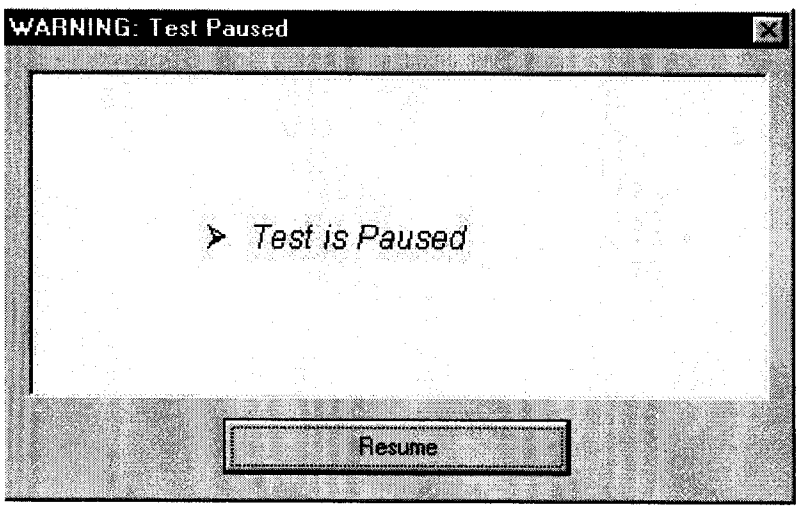
The **ABaer** system displays data about the ABR response in two ways. First, it displays a POVR graph. This graph tracks the progress of the POVR over the number of sweeps sets averaged. This value updates after every 256 sweeps. Please refer to the Collection Window below.



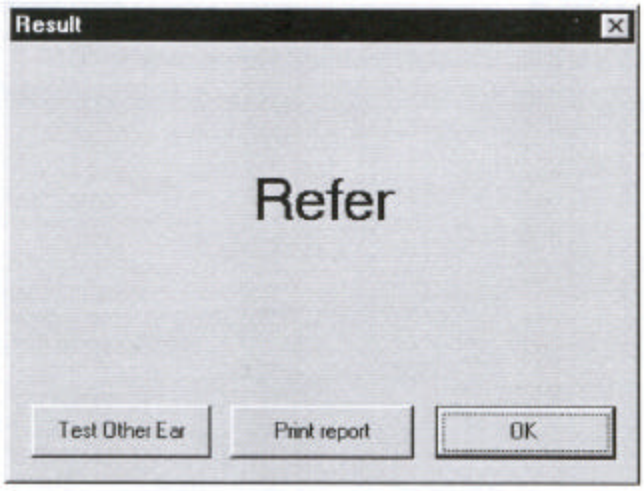
If the setup options allow access to waveform data during collection, selection of the Waveform button in the data collection window displays another window with standard ABR waveform data. Please refer to the Waveforms Window below.



Select Pause to temporarily suspend averaging. The Pause screen will appear as seen below. This is useful if the infant becomes agitated and the screener needs to reposition or calm the baby. Select the Resume button from the Pause reminder screen to resume averaging.

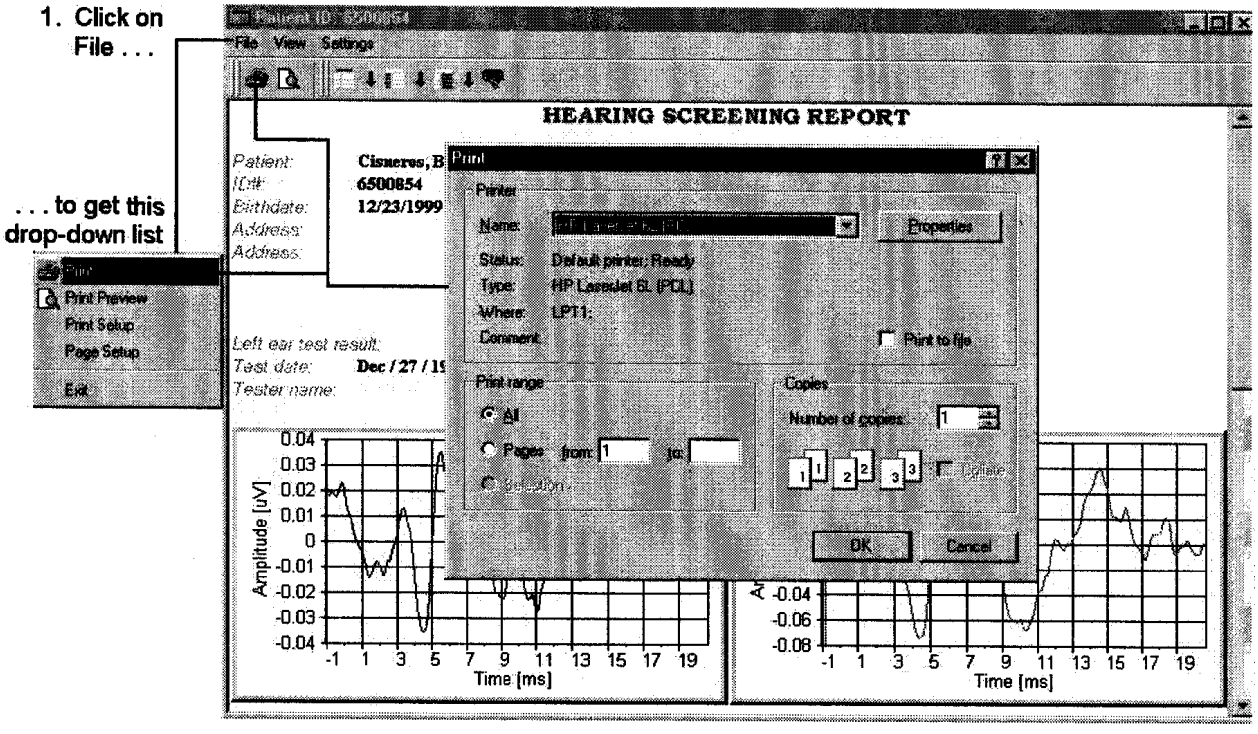


At the conclusion of the test, a display with a PASS or REFER recommendation appears. Please see the Result screen below.



From this screen, the screener may:

- Test the infant's other ear.
  Change the ear probe to the other ear. OR
  Select the Test Other Ear command button.



OR

- Select the Print Report command button.
  This will show a print preview screen. The screener may print a report from that screen.

OR

- Select the OK command button.
  This will return the screener to the **ABaer** Main Screen.

ABaer software provides for many additional features. Please refer to the User's and Service Manual provided by Bio-logic Systems Corp for instruction. This is available from your Regional Audiology Coordinator.

Cleaning and Maintenance

Cleaning with a Mild Antiseptic Solution
Cleanse both cables of the **ABaer** device with a mild antiseptic solution prior to testing the infant. Do not allow fluid to enter the sounds ports of the probe.

Disposable electrode packages as well as disposable ear tips will be provided.

Precautions During a Hearing Screening of an Infant
1. Follow all safety procedures, giving careful and constant attention to the infant and the ABaer equipment.
2. Follow hospital procedures to validate all electrodes are properly applied in order to prevent patient burns while using electro-surgical equipment.
3. Turn the system power on BEFORE you attach the electrodes to the infant. Remove the electrodes from the patient BEFORE you turn the system power off. This avoids shock hazards from power surges.
4. Attach the electrodes to the infant and the system hardware so that the electrode cables can neither entangle or otherwise interfere with the infant.
5. Avoid direct contact between the infant and computer equipment at all times.
6. Do not add any other equipment to the computer.
7. The A**Baer** equipment is not suitable for use in areas of flammable or anesthetic mixture.

Disassembly and Storage
The **ABaer** equipment requires careful handling during disassembly and storage.
1. When a screening is completed, return all controls to their original positions and turn OFF all system equipment.
2. Carefully remove all connecting cables without using force.
3. Follow designated hospital procedures to clean **ABaer** equipment.

Servicing
ABaer equipment requires professional maintenance and repair service. Never attempt to alter, modify or repair this equipment alone.
1. If the **ABaer** equipment is not functioning correctly, mark all defective parts clearly to prevent accidental use before the appropriate repairs can be made.
2. When using a defibrillator, either protect the equipment against defibrillator disharge or remove all patient cables and/or transducers to avoid system damage.
3. Servicing of the **ABaer** is provided by…
4. A qualified Bio-logic service technician should calibrate the **ABaer** system annually to verify calibration of the transducers.

Trouble Shooting

1. Problem: Impedance values are too high and the test will not run.

Solutions:
1) Rescrub the site or sites where indicated. Recall that the impedance value must be below 5.
2) Try using a new electrode or contact pad.
3) Try replacing the entire lead since they occasionally need replacement..

Issues which may arise:

If the impedance values are acceptable and the leads are attached both to the infant and the electrode cable, the ABaer test will run. **To ensure that waveforms are accurate, always check the following:**

1) The probe is in the correct ear. ie. the probe is in the left ear and the ABaer program is actually testing the left ear *or* the probe is in the right ear and the ABaer program is testing the right ear.

2) The electrode leads are correctly attached to the electrode cable – refer to the labels and the colour coding at the connection sites.

Contact your regional coordinator for questions and/or difficulties.

Section 6

In this section you will:

- ✓ Understand how to record the results of a hearing test using the Provincial Tracking System.
- ✓ Initiate a dialogue with parents regarding their infant and the Alberta Universal Newborn Hearing Screening project.
- ✓ Learn who to contact if you have any concerns or difficulties with test results.

Entering the Results of a Hearing Screening Using the Ero-Scan

All the test results will be placed into a database called "Access Ero-Scan 2000".

How To Start The Ero-Scan Database

1. Activate the computer or turn on the computer and wait until the Windows "Desktop" screen appears.

2. Double click on the "Access Ero-Scan 2000" icon to start the program.

3. The following window will appear.

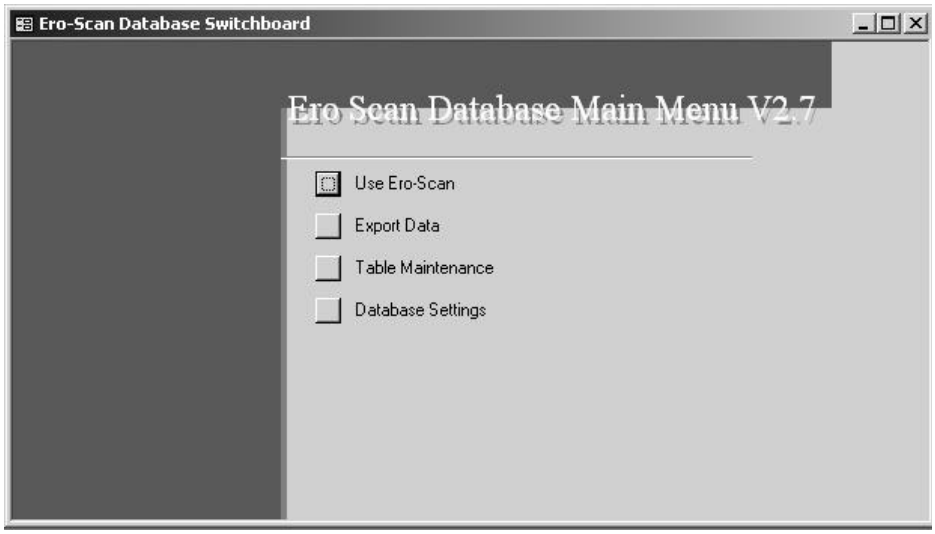


Move the mouse cursor to the text box under "Enter Database Password:". Type in:

baby hear

4. Press Enter

5. The following window will appear. This is the "Ero-Scan Database Main Menu V2.7".



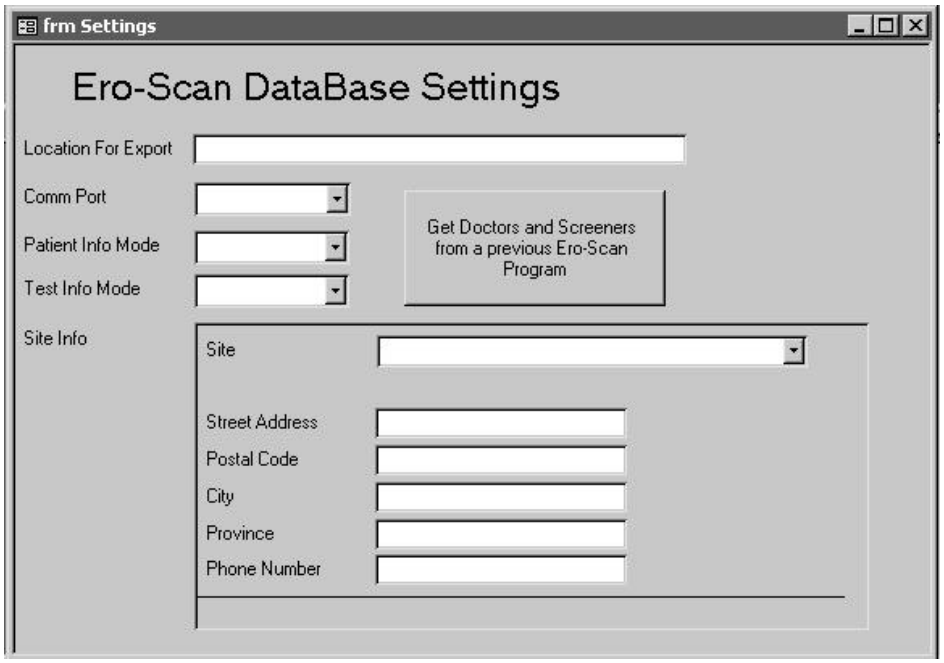
Note: To return to the "Ero-Scan Database Main Menu V2.7" while using Acces Ero-Scan 2000, select (click on) the "C" at the top right hand corner of any window which is currently open.

How To Customize The Database For Your Specific Testing

Inserting The Database Settings

1. On the "Ero-Scan Database Main Menu V2.7", select "Database Settings".

2. The following window will appear.



Enter settings as follows:

Place the mouse cursor in the "Location for Export" field. Type in: *C:\EroScan\Export*

3. Place the mouse cursor in the "Comm Port" field. Type in:
1
This will be the port on the hard drive which the cradle for the Ero-Scan test instrument is plugged into.

OR

Click on the down arrow in the Comm Port field. A drop-down list of Comm Port numbers appears. Select "1" from the list.

4. Place the mouse cursor in the "Patient Info Mode" field. Type in:
Advanced

OR

Click on the down arrow in the Patient Info Mode field. A drop-down list of Patient Info Modes appears. Select "Advanced" from the list.

5. Place the mouse cursor in the "Test Info Mode" field. Type in:
Basic

OR

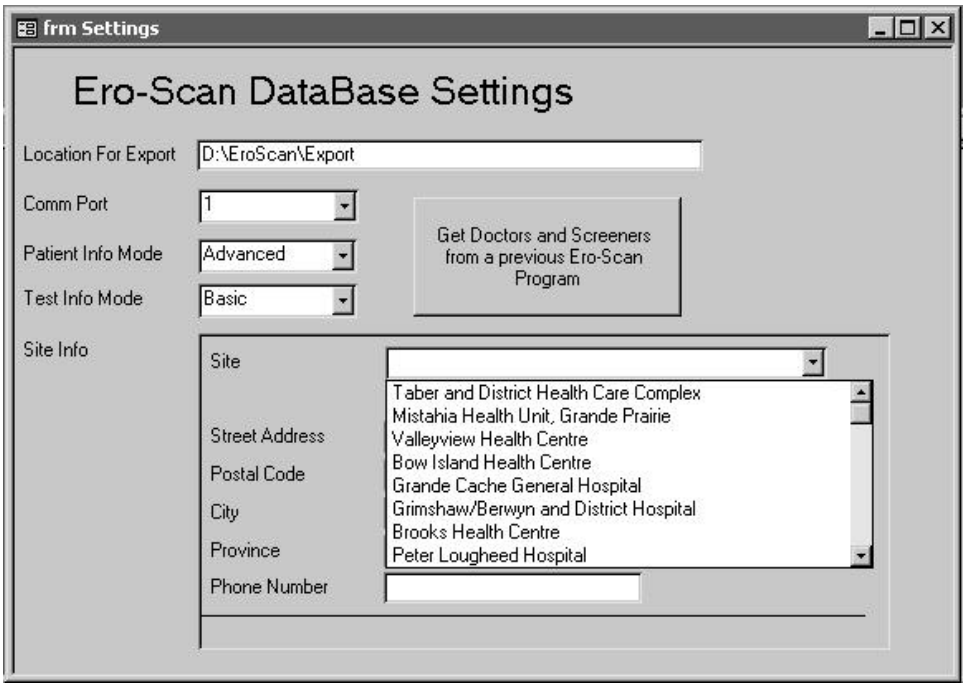
Click on the down arrow in the Test Info Mode field. A drop-down list of Test Info Modes appears. Select "Basic" from the list

6. Select (click on) the "X" at the top right hand corner of the "Database Settings" window.

Setting The Location Of Testing

1. On the "Ero-Scan Database Main Menu V2.7", select "Database Settings".

2. Click on the down arrow in the "Site" field. A drop-down list of sites appears as in the picture below. Select the site at which the test is taking place.



OR

If the location is not present on the list, place the mouse cursor in the "Site" field. Type in the site at which the test is taking place.

3. Place the mouse cursor in the "Street Address" field. Type in the street address of the site.

4. Place the mouse cursor in the "Postal Code" field. Type in the postal code of the site.

5. Place the mouse cursor in the "City" field. Type in the city name of the site.

6. Place the mouse cursor in the "Province" field. Type in the name of the province of the site.

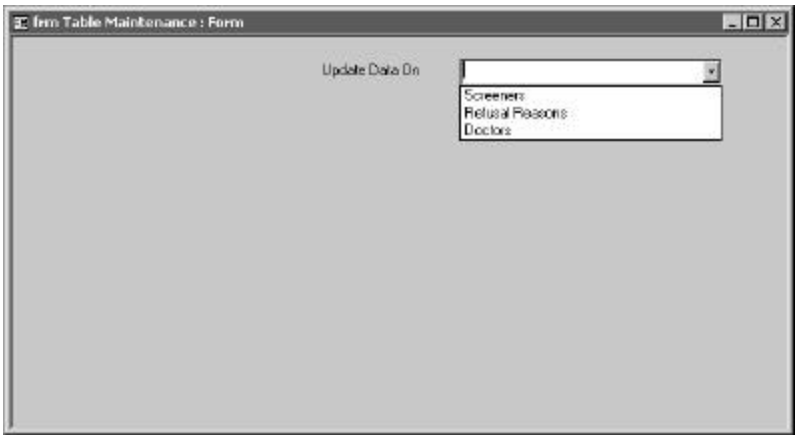
7. Place the mouse cursor in the "Phone Number" field. Type in the phone number of the site.

8. Select (click on) the "X" at the top right hand corner of the "Database Settings" window.

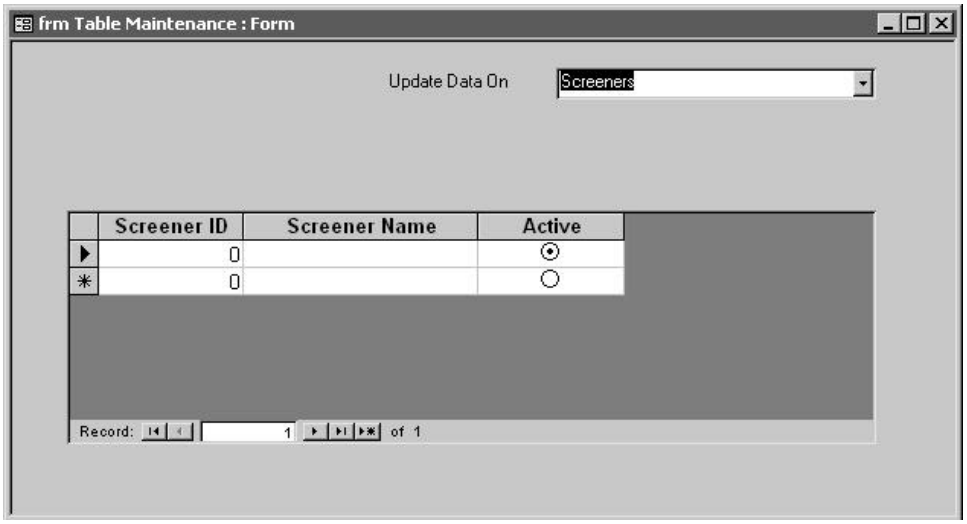
Activating Current Screeners

1. On the "Ero-Scan Database Main Menu V2.7", select "Table Maintenance".

2. Click on the down arrow in the "Update Data On" field. A drop-down list appears as in the picture below. Select "Screeners".



3. The following window will appear.



4. This table includes a list of names of the screeners who run the Ero-Scan tests for your facility. To indicate to the computer which of the screeners on the list is still conducting Ero-Scan tests, activate the current screeners as follows:

   Under the field name "Active," there is a circle which corresponds to the screener name which is found directly to the left of the circle.

   - If the screener name to the left of the circle is conducting Ero-Scan tests at your site, select (click on) the circle so that a black dot appears at the center of it.

   - If the screener name to the left of the circle is not conducting Ero-Scan tests at your site, leave the circle blank.

- If the screener name to the left of the circle was previously active (has a black dot in the circle) but is no longer conducting Ero-Scan tests, select (click on) the circle so that the black dot disappears and the cirlce is left blank.

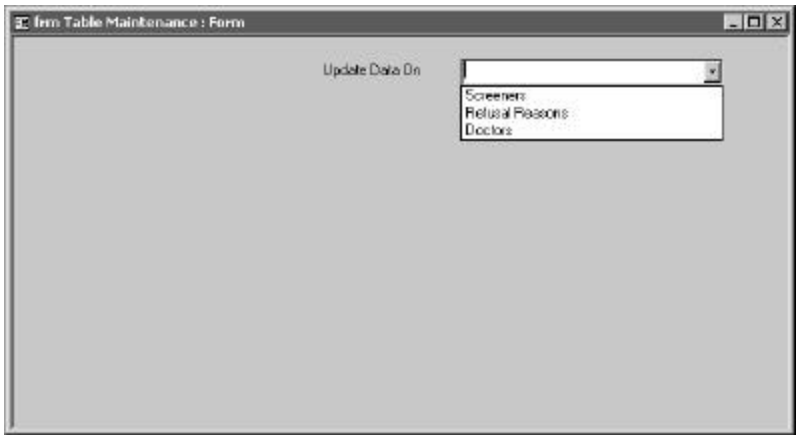
  In the end, the name of every screener at the site which is currently conducting Ero-Scan tests should have a circle filled with a black dot to the right of it.

5. Select (click on) the "X" at the top right hand corner of the "Table Maintenance" window.

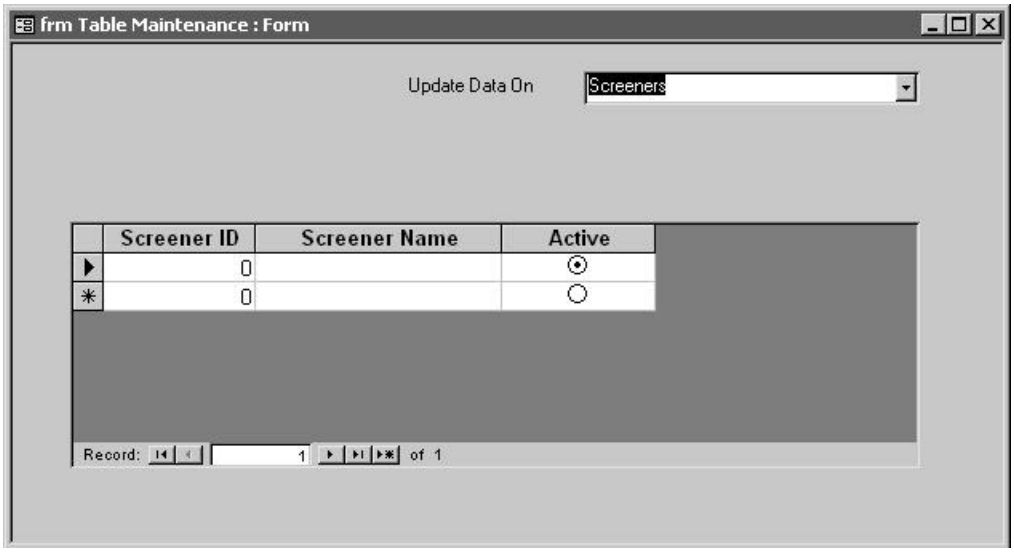
Adding a New Screener To The Database

1. On the "Ero-Scan Database Main Menu V2.7", select "Table Maintenance".

2. Click on the down arrow in the "Update Data On" field. A drop-down list of options appear as in the picture below. Select "Screeners".



3. The following window will appear. This table includes a list of names of the screeners who run the Ero-Scan tests for your facility. Scroll down to the end of this table until there is a blank row in the table.



4. Place the mouse cursor in the left-most empty box in this blank row. In this box, type in the next consecutive number from the box above it. (If there are 9 screeners already in the list and you are adding another then you would type *10* into this box).

5. Press the Tab key on the keyboard. Type in the name of the screener.

OR

Place the mouse cursor in the next box to the right. Type the name of the screener.

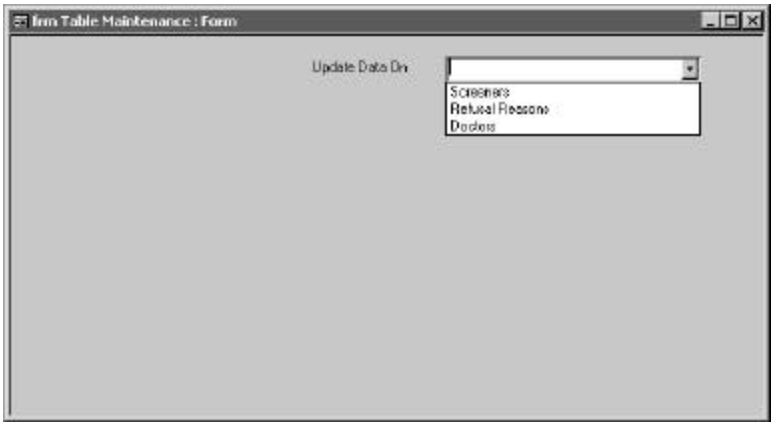
6. Select (click on) the blank circle next to the screener's name you just typed to activate the screener.

7. Select (click on) the "X" at the top right hand corner of the "Table Maintenance" window.

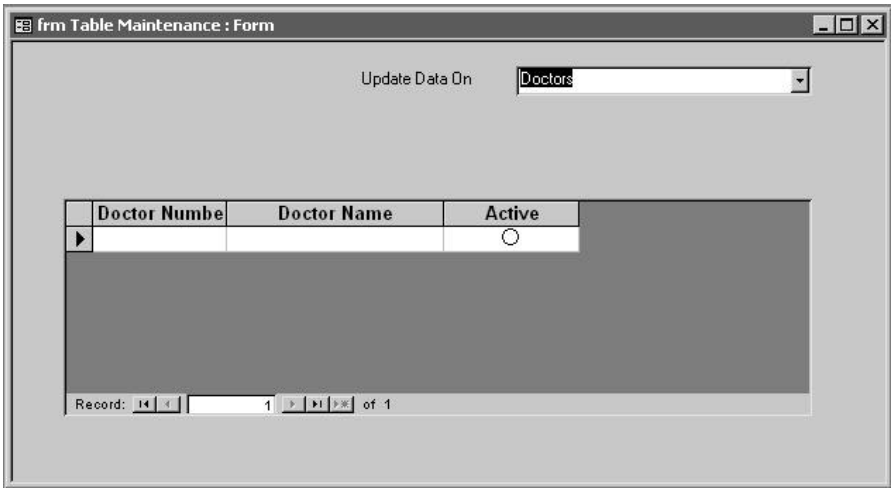
Activating Doctors

1. On the "Ero-Scan Database Main Menu V2.7", select "Table Maintenance".

2. Click on the down arrow in the "Update Data On" field. A drop-down list of options appears as in the picture below. Select "Doctors".



3. The following window will appear.



This table includes a list of names of the doctors who have babies admitted under their care in your institution. To indicate to the computer which of the doctors is admitting infants, activate the current doctors as follows:

Under the field name "Active" there is a circle which corresponds to the doctor's name which is found directly to the left of the circle.

- If the doctor's name to the left of the circle was previously active (has a black dot in the circle) but he/she is no longer admitting patients, select (click on) the circle so that the black dot disappears and the cirlce is left blank.

- If the doctor's name to the left of the circle is admitting infants at your site, select (click on) the circle so that a black dot appears at the center of it.

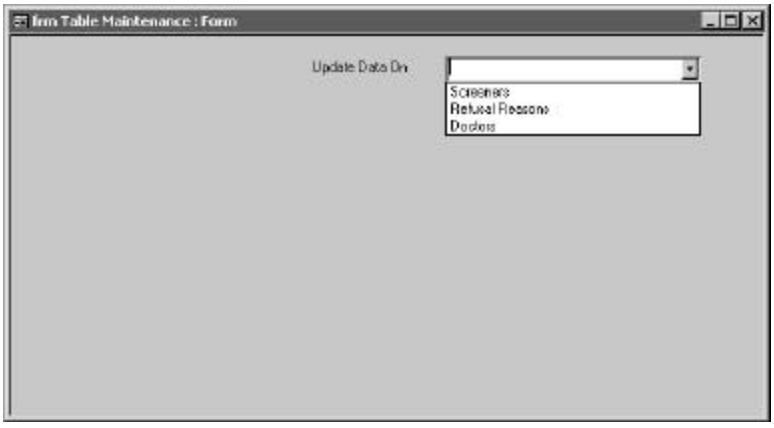
In the end, the name of every doctor at the site who is currently admitting patients should have a circle filled with a black dot to the right of it.

4. Select (click on) the "X" at the top right hand corner of the "Table Maintenance" window.

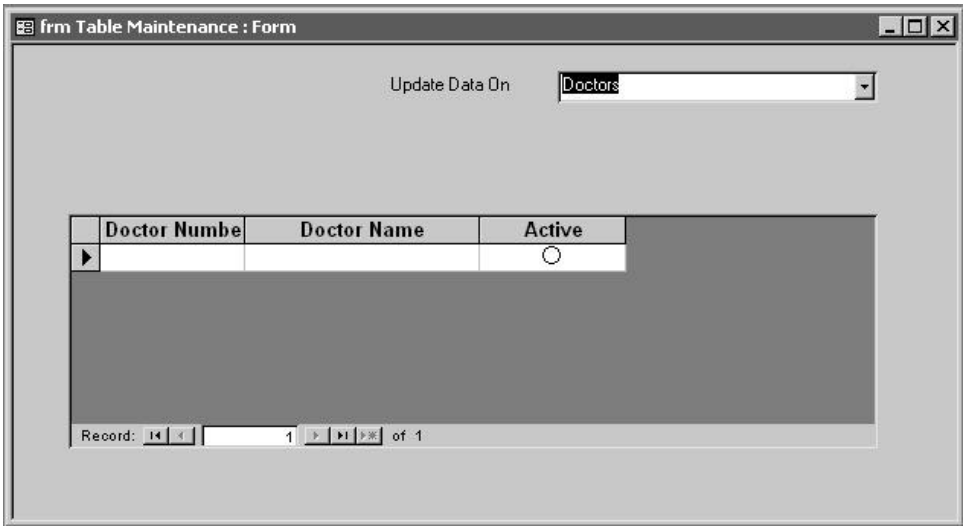
Adding a New Doctor to the Database

1. On the "Ero-Scan Database Main Menu V2.7", select "Table Maintenance".

2. Click on the down arrow in the "Update Data On" field. A drop-down list of options appears as below. Select "Doctors".



3. The following window will appear.



This table includes a list of names of the doctors who have babies admitted under their care in your institution. Scroll down to the end of this table until there is a blank row in the table.

4. Place the mouse cursor in the left-most box in this blank row. In this box, type in the next consecutive number from the box above it. (If there are 9 doctors already in the list and you are adding another, then you would type *10* into this box).

5. Press the Tab key on the keyboard. Type in the name of the doctor.

OR

Place the mouse cursor in the next box to the right. Type the name of the doctor.

6. Select the blank circle next to the doctor's name you just typed to activate the doctor.

7. Select (click on) the "X" at the top right hand corner of the "Table Maintenance" window.

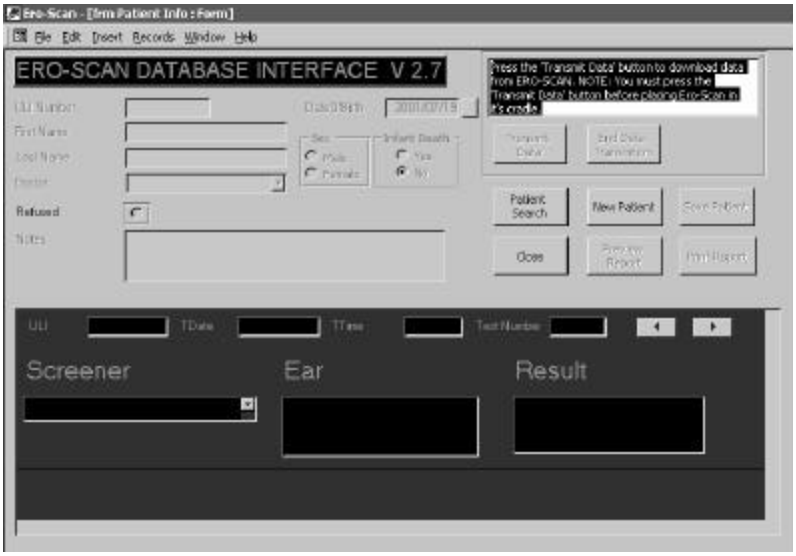
Using The Ero-Scan Database

Inserting New Patient Information Into the Database

1. On the "Ero-Scan Database Main Menu V2.7", select "Use Ero-Scan".

Note: You must insert all patient information before transmitting data.

2. The following screen should appear. This is the interface screen. Select (click on) "New Patient".



3. Enter patient information as follows:

   Place the mouse cursor in the "ULI Number" field. Type in a new ULI number. This is the infant's Unique Identifier Number.

4. Place the mouse cursor in the "First Name" field. Type the infant's first name.

5. Place the mouse cursor in the "Last Name" field. Type the infant's last name.

6. Click on the down arrow in the "Doctor" field. A drop-down list of doctors appears. Select the doctor you want from the list.

7. Place the mouse cursor in the "DateOfBirth" field. Type the infant's date of birth in the YYYY/MM/DD format.

OR

- Click on the box next to the "DateOfBirth" field. A calendar will appear:
- Chose the correct month by clicking on the down arrow next to the name of the month. A drop-down list of months appears. Select the correct month.
- The correct year should appear by default. If you must change the year, click on the down arrow next to the year and select (click on) a year.
- Chose the correct day by selecting (clicking on) the date on the calendar.
- Select (click on) the square box next to the "DateOfBirth" field to enter the adjusted date.



Note: The default date will be the current date of testing.

8. In the “Sex” area, correctly select (click on) the circle next to either the “Male” or “Female” options.

9. In the “Infant Death” area, the “No” option will appear by default. Should the infant die during its hospital stay, select (click on) the circle next to the “Yes” option.

10. If the infant’s parents do not wish to have their infant screened for hearing, select (click on) the circle next to the “Refused” field. A new field will appear named “Refusal Reason”. Click on the drop-down list of refusal reasons and select the correct reason or type in a new reason for the infant’s refusal.

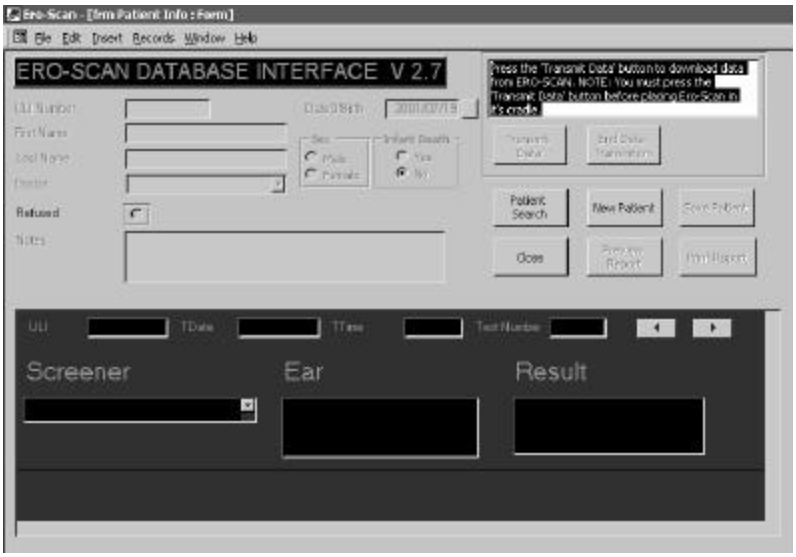
11. If there are any notes to add to the database about the infant’s test, ie. infant congested, place mouse cursor in text box next to the field named “Notes” and type in the note.

12. Select (click on) “Save Patient”. All the entered patient information will now be saved.

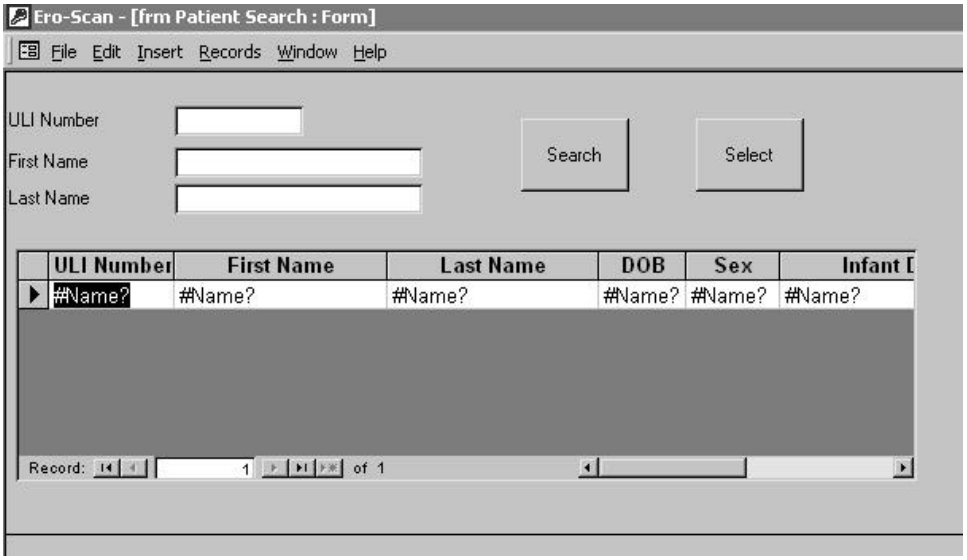
Finding Previously Inserted Patient Information

1. On the “Ero-Scan Database Main Menu V2.7”, select “Use Ero-Scan”.

2. The following screen should appear. Select (click on) “Patient Search”.



3. The following screen will appear.



Place mouse cursor on the "ULI Number" field. Type in the 9- digit ULI number of the infant you are searching for (do not add any spaces or dashes between digits).

4. Place the mouse cursor on the "First Name" field. Type in the first name of the infant you are searching for.

5. Place the mouse cursor on the "Last Name" field. Type in the last name of the infant you are searching for.

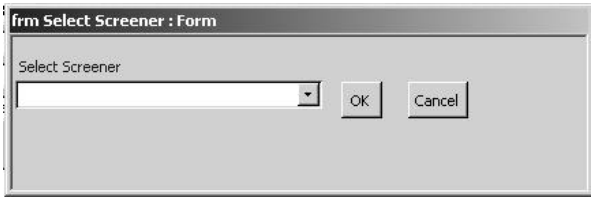
6. Select (click on) "Search".

7. If the correct patient information appears in the table, select (click on) "Select".

Downloading Test Results From Test Instrument to Database

1. Select (click on) "Transmit Data".

2. The following screen will appear:



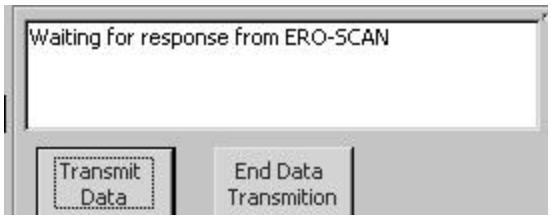
Select (click on) the down arrow next to the field named "Select Screener" and select (click on) the screener's name from the list.

Note: The screener's name selected will be added to the test result for each test done. If a separate screener has done one or more of the tests, this screener's name must be assigned to the test individually after downloading the data from the test instrument.

3. Select (click on) "OK".

4. While holding the test instrument, turn it on by selecting the down button.

5. When the following screen appears:



Place the test instrument into the cradle. The test intrument screen will display the words "Printing Test" and it will turn itself off.

6. To add a different screener to individual tests, select the appropriate test result from selecting the left or right arrows on the top right corner of the test result display screen.

7. Select (click on) the down arrow next to the "Screener" field on the test result display screen and select (click on) the name of the correct screener for this test. The screener's name will be changed for this test only. Repeat steps 6 and 7 as necessary.

Printing A Test Report

1. Once the download is complete, select (click on) "Print Report" from the interface screen.

2. Place the printed copy of the report in the patient's file.

3. Select (click on) the "_" at the top right hand corner of the interface screen.

Exporting Data

1. Select (click on) "Export Data" from the "Ero-Scan Database Main Menu V2.7"

2. The following screen appears saying "Export Complete". Select (click on) "OK". The test results have now successfully been inserted into the database.



The Dialogue with the Parents

As with any other tests or procedures, parents of newborns require a thorough and informed understanding of the hearing screening and why it is being undertaken. Parents should be told exactly what the tests involve and what the results may mean. It is intended that each of the participating regional health authorities in the Alberta UNHS project will provide new parents with literature concerning the project as
well as the implications for their infants. This information should be available to parents at all birthing hospitals.

It is essential to keep in mind that you are dealing with parents of a very new baby and that they may be experiencing some levels of anxiety. Most babies will pass a hearing screening. Of those who do not, their unconfirmed results are often due to very normal and temporary conditions. These include birth debris retained in the ear, the developing immaturity of the baby's central nervous system, the position of the ear canal and most typically, retention of fluid in the middle ear area. Parents sometimes require reassurance. They can be informed that their infant requires a simple rescreening of their hearing. A second screening will typically confirm the infant's hearing.

Infants Diagnosed with a Hearing Loss

The Alberta UNHS project will maintain a record of or "track" those infants in whom normal hearing remains unconfirmed. An UNHS program must involve diagnostic testing to identify the exact nature of the infant's hearing loss. Parents can be encouraged to realize that their infant's hearing loss has been identified well within the first 6 months of its life, a very critical time frame as hearing statistics indicate. The appropriate intervention, whether it be surgical, the teaching of sign language, or most commonly, the use of hearing aids can begin.

Appendix A

Quick Start With the ERO · SCAN DPOAE Test

These steps will allow you to perform a hearing test using the ERO·SCAN device. Please read the entire manual before testing any infant.

Step 1 **Parents are aware that their infant is having a hearing screening.**

Step 2 **The infant is in the appropriate state for a hearing screening and the room is sufficiently quiet to perform the test. Wash your hands and wipe down the ERO·SCAN with a mild antiseptic.**

Step 3 **Select an ear tip for the probe tip of the ERO·SCAN. The ear tip must form a good seal in the ear canal for the hearing test to even begin.**

Step 4 **Turn the ERO·SCAN on by pressing the large ? down arrow button.**

Step 5 **Select the test ear by pressing the ? left or ? right arrow key.**

Step 6 **Gently insert the eartip into the infant's ear canal and obtain a good seal.**

Step 7 **When a seal is obtained, the ERO·SCAN will automatically begin the hearing test by first calibrating and then testing emissions at three f2 frequencies (2, 3 and 4 kHz). The red ERROR light will illuminate if there is noise in the environment. This flashing is normal and will often occur. Once the testing is finished (in approximately 6 seconds), the ERO·SCAN will display "PASS" or "REFER" on the front display.**

Step 8 **If a retest of the initial ear is required, select that ear by pressing the? left or ? right arrow key and repeat steps #6 and #7 to retest this ear.**

Step 9 **When the hearing screening is completed on both ears, enter the results in the designated manner.**

Cleaning:
The probe, cable and housing may be disinfected and/or cleaned with mild agents. Keep all fluids out of the probe and preamp housings. Do not attempt to clean the sound ports on the probe tip. The probe tip is disposable. Replace the tip when it becomes clogged or dirty.

Cautions:
Do not drop or cause hard impact.
Do not insert the probe into an ear canal without an ear tip in place.
Store the **ERO·SCAN** and the remote probe indoors in a cool, dry environment.
If damage is suspected, contact the Regional Audiology Coordinator for repair and/or recalibration of the **ERO·SCAN.**

APPENDIX B

Quick Start With the ABaer (Automated Brainstem auditory evoked response) System

These steps will allow you to perform a hearing test using the ABaer
Please read the entire manual before testing any infant.

Step 1 **Parents are aware that their infant is having a hearing screening.**

Step 2 **The infant is in the appropriate state for a hearing screening and the area is free of any electronic devices capable of causing significant electrical interference ie. portable x-ray machines. Wash your hands and wipe down the two cables of the ABaer device with a mild antiseptic.**

Step 3 **Turn the ABaer system on BEFORE you attach electrodes to the infant.**

Step 4 **Prepare the infant's skin for electrode placement. Swipe the 3 sites gently but firmly with Nuprep. Correctly apply the electrodes to the infant's skin and then connect the leads to their appropriate sites on the electrode cable.**

Step 5 **Select an appropriately sized ear tip for the infant's ear and attach the eartip to the probe on the patient cable. Using the proper technique, gently insert the probe with the attached eartip into the infant's ear canal and obtain a good seal.**

Step 6 **Enter patient information on the first screen of the ABaer program. Select the Accept button.**

Step 7 **Select Right or Left ear button.**

Step 8 **Select the Begin Test button. If the impedance values for all three electrodes are within the preset limits, the data collection phase will automatically begin. Rescrub if impedance values are too high.**

Step 9 **A PASS or REFER recommendation will appear at the test's conclusion. To test the second ear, gently insert the probe with the attached eartip into the other ear and select the Test Other Ear button.**

Step 10 **Following the completion of the second test, select the OK button to return to the ABaer's main screen. Perform any required retests.**

Step 11 **Select Print on the ABaer's main screen to print a report of the test results.**

Step 12 **Gently remove the patient cable and discard the ear tip. Gently remove the electrodes from the infant's skin and discard the electrodes.**

Step 13 **Return all controls to their original positions and turn OFF the ABaer equipment.**

Step 14 **Enter the results of the infant's hearing screening and inform parents of the outcome.**

APPENDIX C

A Review for Screeners of Infant Hearing and Their Role in the Alberta UNHS Project

You may access your understanding the contents of this manual by reviewing the following questions:

1) Universal newborn hearing screening programs are designed to:
a) Identify those infants who have a potential for hearing loss.
b) Ensure that infants with a hearing loss are diagnosed by 3 months of age.
c) Ensure that infants with a hearing loss receive the appropriate treatment by 6 months of age.
d) All of the above.

2) The average age of detection of hearing loss in North America is _____ of age.
a) 3 months
b) 6 months
c) 3 years
d) 5 years

3) The most significant factor which may suggest that an infant is at risk for hearing loss is:
a) A birth weight of less than 1500 grams.
b) The presence of cranio-facial abnormalities such as ear tags or atresias.
c) A strong family history of infant or early childhood hearing loss.
d) Congenital or prenatal infections including Cytomegalovirus, Syphilis, Herpes Simplex virus, Rubella and HIV.

4) Otoacoustic emission tests, such as the DPOAE, work in the following manner:
a) Measure the integrity of outer hair cells in the cochlea or inner ear.
b) Measure the electrophysiologic response of the auditory nervous system to sound.

5) It is not recommended to perform an AABR under these conditions:
a) An x-ray machine is being used in the NICU where you intend to perform a hearing screening.
b) The infant has a gestational age of 32 weeks.
c) The infant is grimacing and making sucking sounds in its sleep.
d) All of the above.

6) The following identification number is used to enter the results of an infant's hearing screening:
a) The mother's hospital admission number.
b) The infant's hospital admission number.
c) The infant's Unique Lifetime Identifier number (ULI).

7) If an infant receives a "REFER" result on his/her DPOAE test, you should inform the parents that:
a) Their infant has a hearing loss.
b) A "REFER" result is an abnormal result and that a retest is required.
c) A "REFER" result is often due to temporary conditions and that a retest is required.
d) They need to clean out their infant's ears.

8) The proper placement of electrodes for the AABR is as follows:
a) The middle forehead and below each ear.
b) The middle forehead and on each ear.
c) The upper shoulder and behind each ear.
d) The middle forehead and on the mastoid bone behind each ear.

9) According to the protocol of the Alberta UNHS Project, an infant will be seen by an audiologist and undergo diagnosis of his/her hearing loss when:
a) The infant obtains a "REFER" response on their first DPOAE test.
b) The infant has unconfirmed hearing on their first AABR test.
c) The infant has unconfirmed hearing at a second rescreening.
d) The mother reports that her one day old infant is not moving or "startling" to sound.

10) Otitis media is the most common childhood disease between birth and age 7 years.
a) True
b) False

References

1. Yoshinaga-Itano C, Sedey AL, Coulter DK, Mehl AL. Language of Early- and later-identified children with hearing loss. Pediatrics 1999; 102(5), pp. 1161-1171.

2. Nicolosi, Lucille, Harryman, Elizabeth, Kresheck, Janet, Terminology of Communication Disorders Speech Language and Hearing. 1978, Waverly Press, Inc. United States, p.96.

3. Brown D.K., Dort J.C. & Sauve R., Newborn Hearing Screening Programs – A truly Canadian Perspective. Journal of Speech-Language Pathology and Audiology. 2001, 24(2), pp.48-58.

Acknowledgements

The Alberta UNHS Project gratefully acknowledges the following in the production of this manual.

Melanie Aikenhead, Calgary, Alberta for her illustration of the infant ear.

Bio-logic Systems Corp, One Bio-logic Plaza, Mundelein, Ill 60060-3700, 1-800-323-8326, Local Phone (847) 949-5200 FAX: (847) 949-8615 for use of diagrams from the **"ABaer** (**A**utomated **B**rainstem **a**uditory **e**voked **r**esponse) System – User's and Service Manual."

Etymotic Research – Maico Diagnostics, 9675 W. 76th Street, Eden Prairie, MN 55344, 612-941-4200, Fax 612-903-4200 for use of illustrations and diagrams from the **"ERO · SCAN** DPOAE Test System User Manual."

The Alberta UNHS Project encourages your comments or questions. If you would like additional copies of this manual or wish to make any recommendations, please contact your

Regional Audiology Coordinator

or

Alberta Universal Newborn Hearing Screening Project
Rm. 193, 3330 Hospital Dr. N.W.
Calgary, AB
T2N 4N1

Contact Numbers:
Tel: (403) 220-4347 Fax: (403) 220-2632
email: babyhear@ucalgary.ca